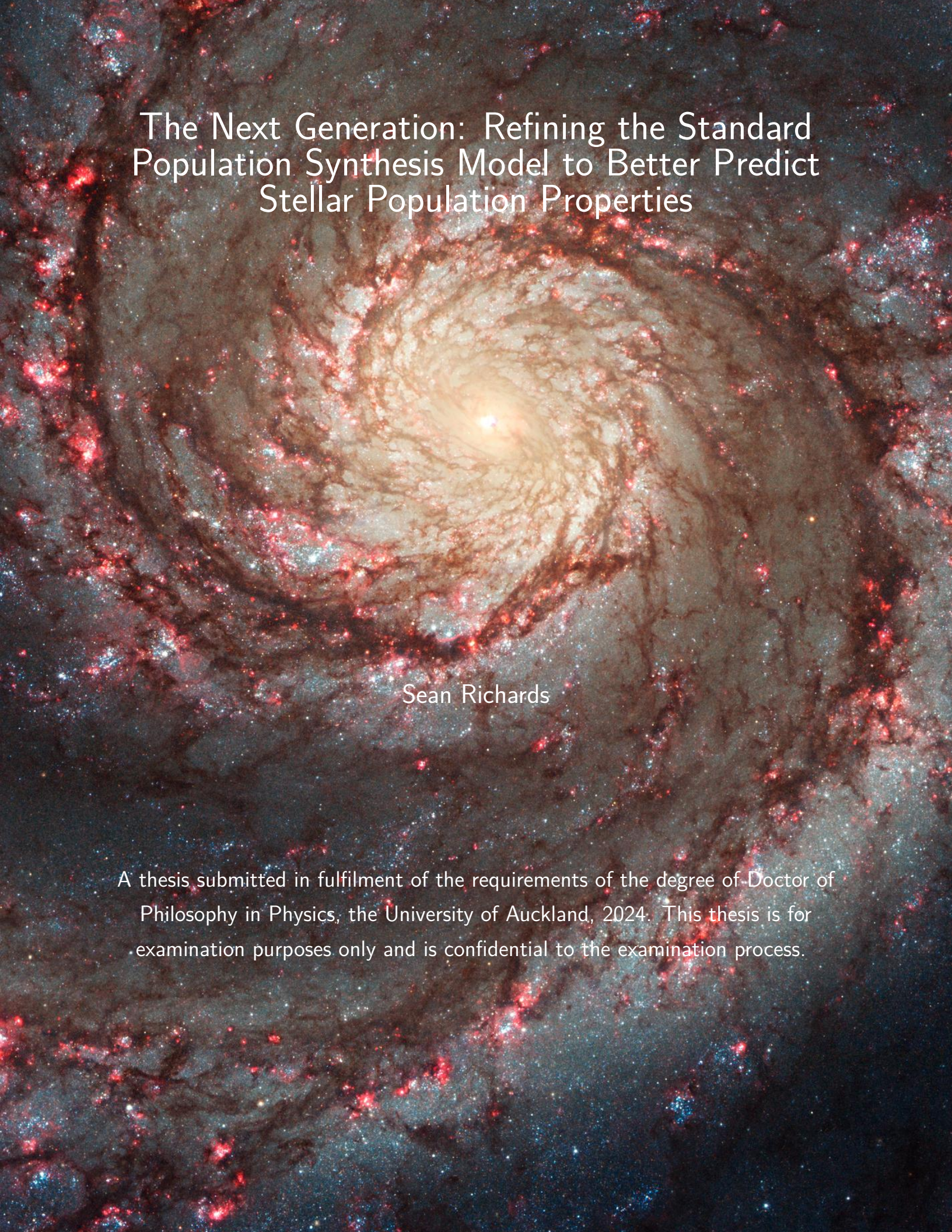




The Next Generation: Refining the Standard Population Synthesis Model to Better Predict Stellar Population Properties

Sean Richards

A thesis submitted in fulfilment of the requirements of the degree of Doctor of
Philosophy in Physics, the University of Auckland, 2024. This thesis is for
examination purposes only and is confidential to the examination process.

Abstract

A short description of the work.

Cover Image: The Whirlpool Galaxy (M51). Image is by NASA and released into the public domain:

https://images.nasa.gov/details/GSFC_20171208_Archive_e001925

For my grandparents: Alan and Nanette, and Walter and Eileen.

Swing low, sweet chariot.

That you are here – that life exists and identity,
That the powerful play goes on, and you may contribute a verse.

– Walt Whitman, *O me! O life!*

Acknowledgements

Ho there, wanderer! Stay thy course a moment and indulge an old man.

ELMINSTER AUMAR

Waipapa Taumata Rau (The University of Auckland) sits within the rohe of several Māori iwi/hapū, including Ngāti Whātua, Ngāti Whātua o Ōrākei, Te Kawerau a Maki, Ngāti Tamaoho, Te Ākitai Waiohū, Ngāti Maru, Te Patukirikiri, Ngāti Pāoa, Ngāti Tamaterā, Ngāi Tai ki Tāmaki, Ngāti Te Ata, and Ngāti Whanaunga, and Waikato-Tainui. Ipipiri (the Bay of Islands) sits within the rohe of Ngāpuhi and Ngāti Rēhia. I acknowledge these iwi and hapū as the tāngata whenua of Tāmaki Makaurau, Ipipiri, and Aotearoa, I acknowledge that sovereignty was never ceded under Te Tiriti O Waitangi, and I pay my respects to kuia and kaumātua of these iwi and hapū, past, present, and future. Tūturu whakamaui, kia tina! Tihei mauri ora!

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This work made use of v2.2 of the Binary Population and Spectral Synthesis (BPASS) models as described in Eldridge et al. (2017) and Stanway & Eldridge (2018).

Once a king or queen in Narnia, always a king or queen.

ASLAN

I am the proud grandchild of immigrants, and by my count, the first person who identifies as Sāmoan to graduate with a PhD from the Department of Physics at the University of Auckland in (at least) 23 years. I’m also one of the few gender non-conforming students in my Department. To all my fellow BIPOC students, all my queer whānau, and all my other underrepresented students in the sciences, I see you, I recognise you, I feel you. Kia kaha, kia māia, kia manawanui.

Who am I?

I am a girl who loves my island

And the girl who loves the sea, it calls me

I am the daughter of the village chief

We are descended from voyagers

Who found their way across the world

They call me

I Am Moana FROM “MOANA”

It would be impossible for me to list my entire whakapapa. It is known as far back as nine generations in parts, tracing back to around the mid-1700s – including to the first Lord Mayor of the City of Sheffield, a

Prime Minister of Samoa, a former Minister of Finance and Chair of Samoa's National Bank, and some of the major clans in Samoa. At last count we know of about 400 relatives – my Great-Great-Aunt's father-in-law had thirteen children in the mid-1800s. But I want to especially mihi to my mother Sue, my late aunt Leslie, my late grandfather Alan, my grandmother Nanette and her late mother Mary "Sugalu" Scott, my aunt Alison and uncle Pete, my great-uncle Ben and great-aunt Lulu, my cousins Jeremy, Jaime, Charlotte, Candice, Ngareta, Wairaka, Marihi, and Tama. To the rest of my 'aiga: e lautele le vateatea, ae le o le avanoa iinei: Ou te alofa ia te outou uma.

Life before death. Strength before weakness. Journey before destination.

FIRST IDEAL OF THE KNIGHTS RADIANT

For Jaffe, Bailey, Riegel, Ray, Mercer, Johnson, Willingham, and O'Brien: Vox Machina, the Mighty Nein, Bells Hells, Gilmore, Allura, Delilah, Pumat, Xandis, and Ryn. For Iyengar, Carazo, Ray, Riegel, Willingham, Wilson, and Mulligan: the Ring of Brass, Vespene, Maya, and Kir. For Lewis: Peter, Edmund, Susan, Lucy, Aslan and Jadis. For Sanderson: Kaladin, Syl, Shallan, Jasnah, Vin, Elend, Kelsier, Rysn, and Hoid. For Vincke, Wilde, English, Newbon, Downie, Solomon, Béart, Jones, Wiles, Mercer, Gregory, Simmons, Isaacs, and Robertson: Lae'zel, Shadowheart, Astarion, Gale, Wyll, Karlach, Halsin, Jaheira, Minsc, Minthara, Ketheric, Gortash, and Orin the Red. For those who imagine the strange and beautiful and complex and weird and mystical worlds of fantasy, whether betwixt the pages of a book, on the silver screen, or upon the roll of a d20, I pay my enormous regards to you.

Can you imagine a piece of the universe

More fit for princes and kings?

I'll trade you ten of your cities

For Marion Bridge and the pleasure it brings

Song for the Mira BY JOHN McDERMOTT

It occurs to me that there are precious few moments in one's life to acknowledge and thank the people who contributed to where one is now. I hold the belief that a person is defined by the sum of their actions and interactions, so I would like to acknowledge a few of the people that have made me the person I am today.

I'd like at this point to acknowledge friends: Connor, Frazer, Girish, Hunter, Pierce, Ryan, Scott, and Sean, with whom I shared some of my first physics and mathematics courses. Emma, the self-appointed creative director of this thesis. Josh, Tessa, Hamish, Jasmine, Ryan: my first experience with D&D was with Eldrin. Andrew, George, Hamish, Connor, Callum, and Mark: Katerina was my longest player experience; playing her taught me so much about myself (and Andrew). Josh, Liam, Arthur, Connor, Andrew, Oliver: Ayo and the rest of the Blue Originators (you never learned their party's name) owe you a debt of gratitude. Even good old Necromancer Bob says hi.

I've lost touch with some of the people I did high school with but the formative experiences there still sit inscribed in indelible ink in my mind. Caitlin, who has been one of my best friends for the better part of a quarter-century (and who would *not* let me live it down were I to forget her). Callum, Susannah, Kyle, Grayson, Tomek "if you're going to try to pronounce my last name, put some **EMPHASIS** on it" Pietkiewicz and the entire business and commerce crew. Sam, Louise, Crooks, the IT Ninjas. Isabella, Chyna, Evan,

Parth, Jay, Matt, all the people at NZSDC 2015, even if I had to argue in favour of alternative medicine and exclusive schools for LGBTQIA youth, those moments will be with me forever. Finn and Susannah: I know they said they never ranked us in maths classes, but they ranked us in maths classes, and I know we always vied for the top three places. Finn is an architect now and Susannah is admitted to the High Court – does that mean I finally win the ranking?

I'd also like to thank Di Eastwood, Maddy Haydock, Melissa Johansson, and Lauren Mulcahy (our debating coach, and my Year 13, 12, and 11 English teachers). Not all of you are teaching anymore, but I still remember vivid parts of your lessons. Miss Mulcahy commending my essay structure (for not opening an essay with "The question I am answering is..." but actually motivating why I enjoyed *To Kill a Mockingbird* and *Romeo+Juliet*). Miss Johansson, for showing faith in the process of my creative writing portfolio ("Sean, I want to send you to counselling for this story. Change nothing, this is an E."). Mrs Haydock, firstly for Mufasa but also for believing in my weird and experimentalist poetry (I was going through a bit of a fourth-wall breaking phase with poetry), for letting me rant for 30 minutes on why *The Matrix* is actually a Marxist allegory and the (concerningly frequent) themes from Freudian psychoanalysis we can pull from it. Moreover, I remember in English class, you congratulating me for using "moreover" correctly in a sentence. Mrs Eastwood for always challenging me to do better in debating. I've said it before and I will say it again: I credit a mix of you four for my erudition and eloquence.

On the topic of teachers, I would like to mihi to Robbie Lumkong, Andy Dyson, Chris Whitaker. Physics lessons in Year 12 and 13 with Mr Lumkong (and in Year 13 with Mr Dyson) were absolutely batshit crazy – can I say batshit in a thesis? – but it is the undeniable infective enthusiasm for physics that someone who has played international rugby¹ brings that made me even think about doing PHYSICS 120 when I came to university and had a free paper in first year. Mr Whitaker, I still remember sitting on a bus behind you in Year 11 Physical Science and asking you a tonne of questions about the big bang and the standard model.

Right, I'm almost done here, I swear.

Physics is a collaborative endeavor, and I would be remiss to not acknowledge my research compatriots. Thank you to the BPASS team, both in Auckland and abroad: John, Heloise, Max, Sohan, Wouter, Petra, Nicolas, Adib, Conor, Gareth, Ashley, and Ankur. We've also been joined by a few students who have since left the team (because they graduated!): Gleb, Emma, Louis, Ethan, Siris, Celina, and Amanda. Thank you to our PIs, Dr Elizabeth Stanway and Prof. Jan Eldridge.

Speaking of Jan! Thank you to Prof. Jan Eldridge and Assoc. Prof. Nick Rattenbury. I genuinely cannot image a more perfect supervision team. Both carry a large amount of mana, and I have learned an immense amount from each of them. You'll notice that all of my contractions of people's titles either omit (in the case of "Dr") or contain periods (in the case of basically every other title). Nick was very pedantic about this in my Masters thesis and now I subconsciously check for this when reading other people's prose. Thank you by extension to Prof. Richard Easter. Jan, Nick, and Richard were my lecturers for PHYSICS 107/G in Semester Two of 2016. I needed extra papers to fill out my first year, as there were only three required papers for my (at that time) Computer Science degree. I took PHYSICS 120 and 150 because I was good at it in High School (see Mr Lumkong, Mr Dyson, and Mr Whitaker above), and PHYSICS 107/G because it sounded fun. I left that course a physics double major, and no joke, restructured my entire degree around taking PHYSICS 356. Jan, Nick, and Richard are a large part of why I am a scientist today. I cannot thank

¹This is not a joke or a quip, my Year 13 physics teacher played as a number eight in the 1999 Rugby World Cup for the USA.

them enough for that. From the bottom of my heart, thank you, ngā mihi, vinaka vakalevu, fa’afetai lava.

I’d also like to mihi to a few of my inspirational heroes in science as a whole. Prof. Stephen Hawking, Prof. Kip Thorne, Prof. Brian Greene, Prof. Brian Cox, Dr Derek Miller (Veritasium), Mrs Dianna Cowern (PhysicsGirl), Mr Destin Sandlin (SmarterEveryDay), Mr John D. Boswell (melodysheep).

*You’ll be in my heart.
No matter what they say.
You’ll be here in my heart.
Always.*

You’ll be in my heart BY PHIL COLLINS

When I started this PhD, I had two alive grandparents. I now have only one, my grandmother Nanette. For Alan, Walter, and Eileen, I carry your memories with me always. In the words of Liam O’Brien: *I was richer for having known you, friend.*

I could not be more keen, for some plain old boring precedented times.

CR. RICHARD HILLS

I’ve noticed that the more theses I write, the more people I thank. My PGDipSci dissertation thanked five people. My Masters thesis thanked some fifty-five, a Government, and several coffee companies. Did I beat it this time?

Run, you clever [girl], and be a Doctor.

CLARA OSWIN OSWALD, *Doctor Who*

*No, thank YOU. One thing about trains... it
doesn’t matter where they’re going. What matters
is deciding to get on.*

THE CONDUCTOR, *The Polar Express*

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Glossary

Notation	Description
pp -chain	proton-proton chain.
$pp1$ -chain	path 1 proton-proton chain.
$pp2$ -chain	path 2 proton-proton chain.
$pp3$ -chain	path 3 proton-proton chain.
3D MHD	3D Magnetohydrodynamics.
AGB	Asymptotic Giant Branch.
BH	Black Hole.
BNS	Binary Neutron Star.
BPASS	Binary Population and Spectral Synthesis.
CCSN	core-collapse supernovae.
CDF	Cumulative Distribution Function.
CME	coronal mass ejection.
CO core	Carbon-Oxygen core.
DTD	Delay-Time Distribution.
GR	General Relativity.
IMF	Initial Mass Function.
ISM	Interstellar Medium.

Notation	Description
JWST	James Webb Space Telescope.
K-S	Kolmogorov-Smirnov.
LIGO	Laser Interferometer Gravitational Wave Observatory.
LISA	Laser Interferometer Space Antenna.
LVK Collaboration	LIGO-Virgo-KAGRA Collaboration.
MLT	Mixing Length Theory.
MWEG	Milky-Way Equivalent Galaxy.
NS	Neutron Star.
NS-BH	Neutron Star - Black Hole Binary.
O3	Operating Run 3.
O4	Operating Run 4.
ONe core	Oxygen-Neon core.
PDF	Probability Density Function.
SFH	Stellar Formation History.
SFRD	Star Formation Rate Density.
SPH	Smoothed-particle hydrodynamics.
USSN	ultra-stripped supernova.
WD	White Dwarf.

Notation	Description
ZAMS	Zero-Age Main Sequence.

Thesis To-do List

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2.3 if this bit stays, 92% of the sun being hydrogen and 0.2% being iron needs a source	10
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2.5 this is a copy-n-paste from the He Stars paper – I need to rewrite it	16
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Thesis Epigraphs

If the stars should appear one night in a thousand years, how would men believe and adore; and preserve for many generations the remembrance of the city of God which had been shown! But every night come out these envoys of beauty, and light the universe with their admonishing smile. (<i>Nature</i> , by Ralph Waldo Emerson)	2
Come with me, and you'll be, in a world of pure imagination. (<i>Pure Imagination</i> , by Gene Wilder)	5
Se'i lua'i lou le 'ulu taumamao. (Pick the breadfruits on the far-off branches first) (Sāmoan proverb)	7
Nature and nature's laws lay hid in night: God said, <i>Let Newton be!</i> and all was light. (Alexander Pope)	7
It did not last: the Devil howling "Ho! Let Einstein be!" restored the status quo. (<i>Answer to Pope's Epitaph for Sir Isaac Newton</i> , by J. C. Squire)	8
There's a trick you can use in mathematics called "not worrying about it" (Matt Parker)	11
And on the pedestal these words appear: "My name is Ozymandias, king of kings: Look on my works, ye Mighty, and despair!" Nothing beside remains. (<i>Ozymandias</i> , by Percy Bysshe Shelley)	13
Pallida mors æquo pulsat pede pauperum tabernas Regumque turres. Pale death knocks with the same tempo upon the huts of the poor and the towers of kings. (Horace)	13
Algebra's like sheet music. The important thing isn't can you read music, it's can you <i>hear</i> it. Can you hear the music, Robert? (<i>Niels Bohr</i> , in Oppenheimer)	18

Wit nodded.	
“Oh, is this the place where I talk?” Kaladin said.	
“If you wish.”	
“I don’t wish. Get on with the story.”	
(<i>Rhythm of War</i> , by Brandon Sanderson)	24
I think there is a world market for maybe five computers.	
(Thomas Watson, President of IBM, 1943)	26
And through the night, behind the wheel, the mileage clicking west	
I think upon Mackenzie, David Thompson and the rest	
Who cracked the mountain ramparts and did show a path for me	
To race the roaring Fraser to the sea.	
(<i>Northwest Passage</i> , by Stan Rogers)	27
Looking up, he saw a thousand stars glittering in the deep velvet of a night with no moon. He knew	
them all, their stories and their names. He knew them in a familiar way, the way he knew his own	
hands.	
(<i>The Name of the Wind</i> , by Patrick Rothfuss)	34
Begin at the beginning, the King said, very gravely, and go on till you come to the end: then stop.	
(<i>Alice’s Adventures in Wonderland</i> , by Lewis Carroll)	35
I fear not the man who has practiced 10,000 kicks once,	
but I fear the man who has practiced one kick 10,000 times.	
(Bruce Lee)	39
If you torture the data enough, they will confess.	
(Salvatore Vitale)	39
Oh! I have slipped the surly bonds of Earth	
And danced the skies on laughter-silvered wings;	
...	
And, while with silent lifting mind I’ve trod	
The high untrespassed sanctity of space,	
Put out my hand, and touched the face of God.	
(<i>High Flight</i> , by John Magee Jr.)	59
Take-off is optional. Landing is mandatory.	
(First rule of aviation)	59

Though in reviewing the incidents of my administration I am unconscious of intentional error, I am nevertheless too sensible of my defects not to think it probable that I may have committed many errors.

(George Washington) 59

Co-Authorship Form

This form is to accompany the submission of any PhD that contains published or unpublished co-authored work. **Please include one copy of this form for each co-authored work.** Completed forms should be included in all copies of your thesis submitted for examination and library deposit (including digital deposit), following your thesis Acknowledgements. Co-authored works may be included in a thesis if the candidate has written all or the majority of the text and had their contribution confirmed by all co-authors as not less than 65%.

Please indicate the chapter/section/pages of this thesis that are extracted from a co-authored work and give the title and publication details or details of submission of the co-authored work.	
Chapter 5 is based on work from “New constraints on the Bray conservation-of-momentum natal kick model from multiple distinct observations”, published in the Monthly Notices of the Royal Astronomical Society (MNRAS), here: dx.doi.org/10.1093/mnras/stad977	
Nature of contribution by PhD candidate	Wrote majority of the paper, developed most parts of the testing pipeline, later data generation, performed the statistical tests.
Extent of contribution by PhD candidate (%)	90

CO-AUTHORS

Name	Nature of Contribution
Jan Eldridge	Original data generation, guidance on project, wrote clarification on the difference between BPASS primary and secondary models, criticism on manuscript
Max Briel	Developing ideas, code commentary, criticism on manuscript
Heloise Stevance	Developing the test for Ultra-Stripped Supernovae, guidance on project, criticism on manuscript
Reinhold Willcox	Providing single-star pulsar dataset from Willcox+ 2021, criticism on manuscript

Certification by Co-Authors

The undersigned hereby certify that:

- ❖ the above statement correctly reflects the nature and extent of the PhD candidate’s contribution to this work, and the nature of the contribution of each of the co-authors; and
- ❖ that the candidate wrote all or the majority of the text.

Name	Signature	Date
Jan Eldridge		
Max Briel		
Heloise Stevance		
Reinhold Willcox		

Introduction

*Una volta che avrai
Spiccato il volo, deciderai
Sguardo verso il ciel saprai:
Lì a casa il cuore sentirai.*

LEONARDO DA VINCI

*Once you have taken flight
You'll decide
Gaze towards the sky, you'll know that:
That is where your heart will feel at home.*

LEONARDO DA VINCI

On September 14, 2015, the Laser Interferometer Gravitational Wave Observatory (LIGO) Collaboration made the first direct detection of gravitational waves, from the merger of a binary black hole with masses $m_1 = 36^{+5}_{-4} M_\odot$ and $m_2 = 29^{+4}_{-4} M_\odot$ (Abbott et al., 2016). This detection sparked the so-called “gravitational wave era”, with detections in Operating Run 4 (O4) reading a cadence of hundreds per year. These detections afford researchers probes of various physical phenomena. For instance, these merger events are some of the most energetic events in the Universe (GW150914 released over 50 times the energy of all stars in the observable Universe, per Abbott et al. (2016)), and the merger of binary neutron stars are the predominant formation channels for most of the elements heavier than iron (see, among others, Eichler et al., 1989; Arcavi et al., 2017; Pian et al., 2017; Smartt et al., 2017). In addition, the stochastic background of gravitational waves is a direct probe on early-Universe cosmology (Christensen, 2018), though the frequency band that these gravitational waves span is between 10^{-9} and 10^{14} Hz (Christensen, 1992, 2018), making detection of the stochastic background difficult (Agazie et al., 2023; Antoniadis et al., 2023; Reardon et al., 2023; Xu et al., 2023).

Any attempt at understanding gravitational waves must be founded in a strong theoretical understanding of stellar structure and evolution. Without theory, any observation is no more than a curiosity; it is with theoretical underpinning and models that insight can be gained. As most astrophysical sources of gravitational waves (i.e., not the stochastic background) are formed by the merger of two “compact remnants” – black holes, white dwarfs, and neutron stars – one must understand their formation and dynamics, and thus the evolution of progenitor stars into these remnants, as well.

Furthermore, a sound understanding of the evolution of stars has impacts beyond gravitational wave astronomy. As the interiors of stars are complex nuclear furnaces, they are responsible for producing most of the elements in the Universe heavier than Hydrogen (Burbidge et al., 1957). In fact, of the elements lighter than Uranium, the only elements not produced in stellar processes (such as through kilonovae or in supernovae) are technetium, promethium (which both lack stable isotopes), boron, beryllium (which both are produced via cosmic ray fission), and some formation channels of lithium. Stars are also the focal point of planetary systems, and provide the theoretical right conditions for the formation of life.

However, this diversity and complexity of stellar interiors demands a careful and varied approach. One cannot abstract away much of the important physics. Although it is common to use approximations such as spherical symmetry to reduce the dimensionality of the problem, and to only evaluate the equations of stellar structure at a fixed number of locations in this one remaining dimension, chosen sufficiently intelligently, a complex computational effort must be undertaken.

*If the stars should appear one night in a thousand years, how would
men believe and adore; and preserve for many generations the
remembrance of the city of God which had been shown! But every
night come out these envoys of beauty, and light the universe with
their admonishing smile.*

Nature, BY RALPH WALDO EMERSON

Observational astronomy can essentially be traced back to the earliest cultures. Babylonia knew of Halley's comet and tracked observations of it as late as 87 BCE (Stephenson et al., 1985). Records of eclipses in Chinese culture exist as far back as 1600 BCE (Fang, 2015). Knowledge of the movements of stars guided Polynesian wayfarers as they crossed the South Pacific (Buck, 1938; Hamacher, 2022). Even in the arts, the knowledge of the lunar perigee was written into *Othello* by Shakespeare:

It is the very error of the moon;
She comes more near the earth than she was wont,
And makes men mad.

– Shakespeare (ca. 1603)

Knowledge of oceanic tides was posited by Ptolemy, writing in *Tetrabiblos* that

ἀμπώτεις καὶ αἱ παλίρροιαί, καὶ αἱ τῶν
πνευμάτων δὲ τροπαὶ μάλιστα περὶ τὰς τοιαύτας
τῶν φωτῶν κεντρώσεις ἀποτελοῦνται πρὸς οὓς
ἂν τῶν ἀνέμων ἐπὶ τὰ αὐτὰ τὸ πλάτος τῆς
σελήνης τὰς προσνεύσεις ποιοῦμενον
καταλαμβάνηται.

Ptolemy & Proclus (ca. 168)

*For the hour by hour intensifications and
relaxations of the weather vary in response to
such positions of the stars as these, in the
same way that the ebb and flow of the tide
respond to the phases of the moon, and the
changes in the air-currents are brought about
especially at such appearances of the
luminaries at the angles, in the direction of
those winds towards which the latitude of the
moon is found to be inclining.*

Ptolemy & Robbins (1940)

This knowledge flowed through to Dante, who wrote in *Paradiso*:

*E come 'l volger del ciel de la luna
cuopre e discuopre i liti senza posa,
così fa di Fiorenza la Fortuna:*

Alighieri (ca. 1317)

*And, as the turning of the lunar sphere covers
and endlessly uncovers the edges of the shore,
thus does fortune deal with Florence.*

Corbett & Webb (2016)

In religious scripture, celestial navigation has oft been a fixture, such as in Chapter 2 of the Gospel according to Matthew:

2:7 – Then Herod, when he had priuily called the Wise men, enquired of them diligently what time the Starre appeared:

2:8 – And he sent them to Bethlehem, and said, Goe, and search diligently for the yong child, and when ye haue found him, bring me word againe, that I may come and worship him also.

2:9 – When they had heard the King, they departed, and loe, the Starre which they saw in the East, went before them, till it came and stood ouer where the young childe was.

– Various Authors (1611)

Despite this deep cultural connection to the stars, what we would now call Astronomy is a somewhat recent invention. Arguably it arose in the 16th century CE with Copernicus proposing the heliocentric model of the solar system, which was confirmed by Galileo in the 17th century CE (Carroll & Ostlie, 2017) – although Galileo’s claims did result in his being convicted of heresy for 300 years.

The early 20th century saw a veritable renaissance in both the observational and theoretical spheres. On the one hand, developments such as the theories of Special Relativity in 1905 (Einstein, 1905), General Relativity (GR) in 1917 (Einstein, 1917), and the predictions of gravitational waves (Poincaré, 1906; Einstein, 1916) led to theoretical revolutions. On the other hand, the innovations of Payne (1925) in showing that stars are composed of mostly Hydrogen and Helium, Baade & Zwicky (1934) in distinguishing “super-novae” from “common novae”, Hoyle (1946) in developing the foundation of the nuclear reaction of elements heavier than Hydrogen in stars, Salpeter (1955) in developing the Initial Mass Function (IMF), Burbidge et al. (1957) in truly developing the theory of stellar nucleosynthesis, Henyey et al. (1955) and Hayashi (1961) in exploring the pre-main sequence evolution of stars were key to kickstarting entire branches of astronomy such as spectroscopy and computational astrophysics.

The pace of discovery of new and exotic astronomical phenomena has not slowed into the 21st century. Thorne-Żytkow objects – giant stars with Neutron star cores (Thorne & Żytkow, 1977) – once only existed as theoretical curiosities, yet now have credible observational candidates in Levesque et al. (2014). The aforementioned gravitational waves were detected in 2015 (LIGO Scientific Collaboration and Virgo Collaboration, 2016), a century after they were demonstrated by Einstein (1916) to be a central prediction of GR. More recently, Lau et al. (2022) directly imaged 17 carbon-rich dust rings around WR140 with James Webb Space Telescope (JWST), enriching the Interstellar Medium (ISM) with carbon-rich compounds. In the coming years, the Vera C. Rubin Observatory (Ivezić et al., 2019) and Laser Interferometer Space Antenna (LISA)

are expected to vastly increase the number of short-period eclipsing double white dwarfs (Burdge et al., 2020), with the latter able to detect gravitational waves with frequencies between 0.1 mHz and 1 Hz (Amaro-Seoane et al., 2012), compared to the $\sim 10 \text{ Hz} - 1000 \text{ Hz}$ that LIGO-Virgo-KAGRA Collaboration (LVK Collaboration) can detect (Schmitz, 2021). The stochastic gravitational-wave background was detected by Agazie et al. (2023); Antoniadis et al. (2023); Reardon et al. (2023); Xu et al. (2023), via a search of pulsar-timing arrays.

Astronomy has evolved from the era of the naked-eye, to the era of the telescope, to the era of the observatory, and now the gravitational wave era. The next era of astronomy is the big data era. With observations reaching sub-day cadences, a large array of theoretical models, and perhaps the development of “on-demand” detailed models is required.

1.2

Stellar Modelling

Modelling the evolution of a star is a nontrivial endeavour. Stars as objects obey General Relativity (GR) owing to their masses and sizes, and reach internal temperatures, pressures, and densities unsustainable on Earth. For instance, the core of a star is capable of nuclear fusion, which has not been sustainably recreated on Earth. Moreover, the density of Sirius B – a White Dwarf (WD) in the constellation of Canis Major – is $2.14 \times 10^{10} \text{ kg m}^{-3}$ (Barstow et al., 2005) and the mean density of the Sun is $\sim 1410 \text{ kg m}^{-3}$ (Eddy & Ise, 1979). On Earth, the highest density measured in a laboratory is $\sim 27\,000 \text{ kg m}^{-3}$, requiring pressures on the order of 2.3 TPa (Fratanduono et al., 2020), and observed for only nanoseconds.

Because of these factors, one often turns to computer simulations to model the interiors of stars, as it is nigh impossible to achieve and sustain these conditions to experimentally test any physical phenomena. That said, missions such as the Parker Solar Probe [TODO: cite] have made flyby missions into the atmosphere of the Sun, though broadening the sample size to two stars – using Proxima Centauri, the nearest star to Earth aside from the Sun – would require travelling at approximately $0.1c$ for 40 years to reach. If instead we were to send the Parker Solar Probe travelling at a constant velocity of its top velocity of $\sim 176\,471 \text{ m s}^{-1}$, it would take approximately 7200 years to reach Proxima Centauri. Needless to say, physical experimentation with a star is difficult.

1.2.1

Population Synthesis Codes

Broadly speaking, there are three approaches to modelling stars, all of which have both merits and drawbacks. These three methods are:

1. Rapid Population Synthesis,
2. Detailed Population Synthesis, and
3. 3D Magnetohydrodynamics (3D MHD) Simulations.

Each of these approaches informs the other two, and all play important roles in stellar modelling. Although we focus on detailed methods, we will briefly outline all three approaches here.

Detailed Population Synthesis

Detailed stellar models explicitly solve the equations of hydrostatic equilibrium, usually via some form of matrix-inversion method. For instance, as will be discussed in subsection 2.7.1, the Cambridge STARS code inverts the Henyey matrix $\mathbf{H}_{ij} = \frac{\partial(\text{equation } i)}{\partial(\text{variable } j)}$. MESA (Paxton et al., 2011, 2013, 2015, 2018, 2019; Jermyn et al., 2023) uses a Newton-Raphson solver in order to converge to a solution. In particular, we use a variant of the Cambridge STARS code, which solves the equations of *both* stars simultaneously.

Rapid Population Synthesis

Rapid population synthesis methods exploit the statistical properties of stars in order to rapidly infer properties of populations of stars. However, they still require as input detailed population synthesis models – for example, COMPAS (Team COMPAS: Riley et al., 2022) uses MESA (Paxton et al., 2011, 2013, 2015, 2018, 2019; Jermyn et al., 2023) models as its input models.

3D MHD Simulations

3D Magnetohydrodynamics (3D MHD) and Smoothed-particle hydrodynamics (SPH) simulations are the most detailed method of stellar modelling. The base principle is that the star is evaluated as a network of particles in a continuous media, and the evolution of the star is followed in theoretically atomic detail. For example, PHANTOM (Price et al., 2018) is a SPH code which essentially models the star as a fluid, where the Lagrangian hydrodynamical equations are solved for in two-dimensional cells (termed “particles”). Although these present the most detailed models, they are also the most time-intensive to solve. For a SPH code, the famous Navier-Stokes equation is invoked, which has no analytical solution but is computed in excruciating detail. As a result, most SPH models take weeks to run and simulate a star for kiloseconds of its lifetime, rather than for the millions of years detailed models are able to compute.

Though it may seem that they serve the same niche and are therefore competing to be the “better” method of stellar modelling, detailed 1D models and 3D MHD/SPH models exist harmoniously with each other. The advancements in 3D MHD/SPH modelling inform the evolution of 1D models in a process known as “3-to-1”, and the computational flexibility of 1D models offers insights into where 3D models wish to target to evolve.

1.3

Thesis Outline

Come with me, and you'll be, in a world of pure imagination.

Pure Imagination, BY GENE WILDER

This thesis is, broadly, a tour of stellar evolution from Zero-Age Main Sequence (ZAMS) to the eventual merger of two compact objects, with a few omissions along the way (such as the moment of supernova). The thesis is structured as follows. In Chapter 2, we provide the necessary theoretical background for future chapters. In Chapter 3, we critically examine the pieces of software and numerical methods employed in this work, and discuss the logistics of running this single-thread grid-based code as a multiprocessed code on a distributed cluster. In Chapter 4, we present the results of our stellar evolution codes: both as results for single and binary star evolution, as well as the interesting case when both stars almost (or in some instances,

do) touch. In Chapter 5, we examine the kinematics of the post-supernova inspiral and the mechanics of the natal supernova kick. In Chapter 6, we review the thesis and conclude. Finally in the appendices, we provide the KAITIAKI code necessary to generate the models for the stellar evolution plots throughout the thesis.

Theoretical Background

Se'i lua'i lou le 'ulu taumamao. (Pick the breadfruits on the far-off branches first)

SĀMOAN PROVERB

In this chapter, we provide a brief account of the evolution of a star from a theoretical perspective, from its formation to post-supernova. We will discuss both the case of a single isolated star, and an isolated binary. It is pertinent to note that our binaries will be assumed to have formed at birth, rather than by, for example, the dynamical capture of a runaway star in a globular cluster, which introduces extra effects and considerations. This process can be modelled by evolving a single star to a certain point, and then enabling binary evolution.

2.1 Stellar Formation

Nature and nature's laws lay hid in night: God said, Let Newton be! and all was light.

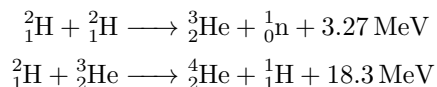
ALEXANDER POPE

From the point of view of stellar evolution, which starts in earnest at the ZAMS, the formation of stars is interesting but ultimately broadly unimportant. We provide a short recap of stellar formation here as scene-setting, but our stellar evolution models assume that a gas cloud has already collapsed and we have a star at its ZAMS.

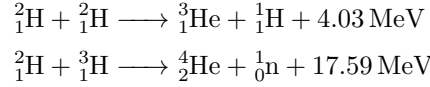
Stars are born in clouds of dust in rich H(II) regions of space. Due to inhomogeneities in the dust formation, some regions become more dense than others. When a region of this gas with adiabatic constant γ and mean molecular weight μ reaches the so-called Jeans Mass,

$$M_J = \left(\frac{10k_B T}{3(\gamma - 1)\mu m_H G} \right)^{\frac{3}{2}} \left(\frac{3}{4\pi\rho} \right)^{\frac{1}{2}} \sim \left(\frac{10k_B T}{G} R \right)^{\frac{3}{2}} \left(\frac{1}{M} \right)^{\frac{1}{2}} \quad (2.1)$$

where the similarity is taking $\gamma = \frac{5}{3}, \mu = \frac{1}{2}$ for an adiabatic gas composed of H(II), then gravity becomes the dominant force in the clumps of gas. This causes a protostar to form. This protostar slowly gains more mass from its surroundings and, as it does so, increases in temperature. When the core temperature reaches 10^7 K, the hydrogen in the core begins to fuse into helium via the reaction chain



This is aided by two further hydrogen to helium reactions:



or, via a master equation,



with intermediate products which remain stable and are thus omitted for simplicity. This energy release will sustain the star throughout its lifetime, providing outwards pressure resisting gravity pulling the star inwards, in a balance termed *hydrostatic equilibrium*. Once hydrogen fusion has begun, we say that the star has reached ZAMS, and is now on the *main sequence*, where it will spend approximately 90% of its lifetime.

2.2

Stellar Evolution

It did not last: the Devil howling "Ho! Let Einstein be!" restored the status quo.

Answer to Pope's Epitaph for Sir Isaac Newton, BY J. C. SQUIRE

Now that our protostar has evolved onto the main sequence, it has become a star in its own right. Over the next few million years, it will continue to burn hydrogen into helium, forming a Helium core. It is at this point that the evolution encounters a number of bifurcation points, depending on the mass and chemical composition of the star. In general, while the star lives, the four equations of stellar structure hold:

$$\left. \begin{aligned} \frac{dm}{dr} &= 4\pi r^2 \rho \\ \frac{dp}{dr} &= -\frac{GM\rho}{r^2} \\ \frac{dT}{dr} &= -\frac{3\kappa\rho L}{64\pi r^2\sigma T^3} \\ \frac{dL}{dr} &= 4\pi r^2 \rho \epsilon \end{aligned} \right\} \quad (2.3)$$

where ϵ is the rate of energy eneration per unit mass, κ is the opacity of the material, σ is the Stefann-Boltzmann constant, and all other symbols have their usual meanings.

These equations represent conservation of mass, hydrostatic equilibrium, energy transport, and thermal equilibrium, respectively. Knowing constants ϵ, κ , we have four equations for four unknowns. It is usually convenient to take the *Rosseland mean* opacity,

$$\frac{1}{\kappa} = \frac{\pi}{acT^3} \int_0^\infty \frac{1}{\kappa_\nu} \frac{\partial B}{\partial T} d\nu \quad (2.4)$$

where B is the black-body intensity at frequency ν and temperature T . The function κ_ν is obtainable from nuclear physics: it is intimately related to the scattering cross-section σ :

$$\frac{1}{\kappa_\nu \rho} = \frac{1}{\sigma n} \quad (2.5)$$

where ρ is the mass density of the star, and n is the particle number density.

On the other hand, ϵ is obtainable from knowing the nuclear reactions occurring in the star's core:

$$\epsilon = \sum_i^N \text{energy output of reaction } i \quad (2.6)$$

All told, in the simplest sense, the internal *structure* of the star can be modelled by hand by solving Equation 2.3 from $m = 0$ to $m = M$.

2.2.1

The chemical composition of stars

A star contains at its core a complex nuclear furnace. Since stars are born in regions of space rich in H(II) – molecular hydrogen – and obtain high enough pressures in the core to ignite fusion, the star will be sustained by the burning of predominantly Hydrogen and Helium. We thus group the three main stages into Hydrogen, Helium, and Metals (which we define as anything with atomic number ≥ 3 , i.e., Lithium and heavier elements). It is convenient to speak of the mass fraction of the element, i.e., the fraction by mass of a star that is a certain element. For our purposes, this is expressible as

$$X + Y + \sum_{i=0}^{115} Z_i = 1, \quad (2.7)$$

where X is the mass fraction of Hydrogen, Y of Helium, and Z_i as the metal¹ with atomic number $i + 3$. When speaking about the chemical composition of a star it is also convenient to speak of the *metallicity*, the fraction at ZAMS of the star that is metal. For this, we define the metallicity as

$$Z \equiv \sum_{i=0}^{115} Z_i, \quad (2.8)$$

such that Equation 2.7 becomes

$$X + Y + Z = 1. \quad (2.9)$$

Hydrogen burning: the *pp*-chain

A star spends $\sim 90\%$ of its lifetime on the main sequence, which is where it burns protons (bare hydrogen atoms) into helium. This reaction is called the proton-proton chain (*pp*-chain), and can be broken up into three potential paths. All three paths begin with the reaction **[TODO: finish]**

1. path 1 proton-proton chain (*pp1*-chain):

2. path 2 proton-proton chain (*pp2*-chain):

¹It is custom in astronomy to refer to anything with an atomic number greater than 2 as a metal, rather than the conventional definition of a material conductive at absolute zero, including most of the d-block, s-block, f-block, and about half of the p-block on the periodic table.

3. path 3 proton-proton chain (*pp3-chain*):

The metallicity of the Sun

It is often convenient to index our metallicities relative to that of the Sun. To do this, we of course need to know the chemical composition of the Sun. We ignore virtually every isotope of every element in favour of $^{56}_{26}\text{Fe}$ in particular. This is for a few reasons: it is the most stable element of fusion (nuclides higher than $^{56}_{26}\text{Fe}$ will undergo fission to it and nuclides lower than $^{56}_{26}\text{Fe}$ will fuse towards it), it is very abundant observationally, and its absorption lines in spectra are easily distinguishable. Thus we can base our metallicity calculations via a quantity known as the iron-abundance ratio,

$$[\text{Fe}/\text{H}] = \log_{10} \left(\frac{N_{\text{Fe}}}{N_{\text{H}}} \right) \Big|_{\text{star}} - \log_{10} \left(\frac{N_{\text{Fe}}}{N_{\text{H}}} \right) \Big|_{\text{sun}}, \quad (2.10)$$

where N_{H} (N_{Fe}) is the number of Hydrogen (Iron) atoms per unit volume of the sun or star. Mechanically this is represented by

$$N_{\text{X}} = N_{\text{A}} \frac{M}{m_{\text{X}}} \quad (2.11)$$

where N_{A} is Avogadro's constant ($\sim 6.022 \times 10^{23} \text{ mol}^{-1}$), M is the mass of element X in the star, and m_{X} is the relevant molar mass. For instance, imagining a hypothetical iron-rich star of mass M consisting of 50% hydrogen and 50% iron, the $[\text{Fe}/\text{H}]$ would be

$$\begin{aligned} [\text{Fe}/\text{H}] &= \log_{10} \left(\frac{N_{\text{A}} \frac{M_{\text{Fe}}}{m_{\text{Fe}}}}{N_{\text{A}} \frac{M_{\text{H}}}{m_{\text{H}}}} \right) \Big|_{\text{star}} - \log_{10} \left(\frac{N_{\text{A}} \frac{M_{\text{Fe}}}{m_{\text{Fe}}}}{N_{\text{A}} \frac{M_{\text{H}}}{m_{\text{H}}}} \right) \Big|_{\text{sun}} \\ &= \log_{10} \left(\frac{M_{\text{Fe}} m_{\text{H}}}{m_{\text{Fe}} M_{\text{H}}} \right) \Big|_{\text{star}} - \log_{10} \left(\frac{M_{\text{Fe}} m_{\text{H}}}{m_{\text{Fe}} M_{\text{H}}} \right) \Big|_{\text{sun}} \\ &= \log_{10} \left(\frac{0.5M m_{\text{H}}}{m_{\text{Fe}} \times 0.5M} \right) \Big|_{\text{star}} - \log_{10} \left(\frac{0.002M_{\odot} m_{\text{H}}}{m_{\text{Fe}} \times 0.92M_{\odot}} \right) \Big|_{\text{sun}} \\ &= \log_{10} \left(\frac{m_{\text{H}}}{m_{\text{Fe}}} \right) \Big|_{\text{star}} - \log_{10} \left(\frac{0.002M_{\odot} m_{\text{H}}}{m_{\text{Fe}} \times 0.92M_{\odot}} \right) \Big|_{\text{sun}} \\ &\sim 2.66 \end{aligned}$$

[TODO: if this bit stays, 92% of the sun being hydrogen and 0.2% being iron needs a source]

In this regime, we can express the metallicity as approximately

$$\mathcal{Z} = 10^{[\text{Fe}/\text{H}]} \mathcal{Z}_{\odot}, \quad (2.12)$$

and thus by knowing the simple iron content of the Sun or star, we can approximate the metallicity. Our iron-rich “star” would thus have a metallicity of approximately 460 times that of the sun.

Asplund et al. (2021) presents an estimate **[TODO: finish sentence]**

In this section, we examine the four major regimes of stellar evolution for single-stars. The precise boundaries are debatable, but are generally accepted to be logarithmically spaced.

Low-Mass Stars: $0.08 M_{\odot} < M < 2 M_{\odot}$

Low-mass stars ignite helium degenerately in their cores, leading to thermal runaway. The star can peak at a luminosity close to $10^{11} L_{\odot}$ (approximately 100 times the peak luminosity of SN 1987A, per Figure 5 of Arnett et al. (1989)). However, this peak in luminosity is attributable to nuclear fusion in the core, and the energy does not reach the surface, let alone escape into the ISM – rather, it is absorbed into the outer layers of the star. During this extremely brief thermal runaway, the luminosity begins to settle and the star is able to continue evolving. Models of low mass stars fail at this stage and care is required to evolve them.

After the ignition of helium, these stars cross the so-called Hertzsprung Gap – the region where very few stars are observed empirically. This region is bounded (on a colour-magnitude diagram, which is a transform of the HR diagram)² by the points $(M_v, B - V) = \{(2, 0.5), (-7, 0.1), (-6, 1.6)\}$ (Hoyle, 1960), where M_v is the V-band magnitude (magnitude in the visual band), and $B - V$ is the colour index of the star. The phenomenon of a gap existing is attributable to the fact that stars cross this gap exceedingly quickly: whereas it spends $\sim 90\%$ of its life on the main sequence, it spends $\sim 2\%$ of its life crossing the Hertzsprung gap (Hoyle, 1960).

Intermediate-Mass Stars: $2 M_{\odot} < M < 8 M_{\odot}$

Massive Stars: $8 M_{\odot} < M < 80 M_{\odot}$

Very Massive Stars: $80 M_{\odot} < M < 300 M_{\odot}$

There's a trick you can use in mathematics called "not worrying about it"

MATT PARKER

This analysis has presumed Newtonian mechanics. However, the mass of stars puts them formally into the general relativistic regime, and thus strictly speaking we should treat this as a problem in full GR. In this approach, we would solve the Einstein Field Equations for the interior of the star:

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu} \quad (2.13)$$

with cosmological constant Λ , stress-energy $T_{\mu\nu}$, and the Einstein tensor $G_{\mu\nu}$ given by

²The relationship between a colour-magnitude diagram and the HR diagram is that $\log L = 0.4 (M_{\text{bol},\odot} - M_{\text{bol},*}) \simeq 0.4 (4.74 - M_{v,*} - \text{BC})$, $\log T = \log \left(4600 \left(\frac{1}{0.92(B-V)+1.7} + \frac{1}{0.92(B-V)+0.62} \right) \right)$, where BC is a bolometric correction which depends on the stellar type. This expression for the temperature is a fit provided by Ballesteros (2012).

$$G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R \quad (2.14)$$

$$= R^\lambda{}_{\mu\lambda\nu} - \frac{1}{2}g_{\mu\nu}R^\sigma{}_\sigma \quad (2.15)$$

$$= R^\lambda{}_{\mu\lambda\nu} - \frac{1}{2}g_{\mu\nu}g^{\sigma\gamma}R_{\sigma\gamma} \quad (2.16)$$

$$= \sum_\lambda R^\lambda{}_{\mu\lambda\nu} - \frac{1}{2}g_{\mu\nu} \sum_\sigma \sum_\gamma \sum_\alpha g^{\sigma\gamma} R^\alpha{}_{\sigma\alpha\gamma} \quad (2.17)$$

$$= \sum_\lambda \left(\Gamma^\lambda{}_{\mu\nu,\lambda} - \Gamma^\lambda{}_{\mu\lambda,\nu} + \sum_\xi \Gamma^\lambda{}_{\xi\lambda} \Gamma^\xi{}_{\mu\nu} - \sum_\xi \Gamma^\lambda{}_{\xi\nu} \Gamma^\xi{}_{\mu\lambda} \right) \\ - \frac{1}{2}g_{\mu\nu} \sum_\sigma \sum_\gamma \sum_\alpha g^{\sigma\gamma} \left(\Gamma^\alpha{}_{\sigma\gamma,\alpha} - \Gamma^\alpha{}_{\sigma\alpha,\gamma} + \sum_\xi \Gamma^\alpha{}_{\xi\alpha} \Gamma^\xi{}_{\sigma\gamma} - \sum_\xi \Gamma^\alpha{}_{\xi\gamma} \Gamma^\xi{}_{\sigma\alpha} \right), \quad (2.18)$$

where there is still tensorial shorthand:

$$\Gamma^\gamma{}_{\beta\mu} \equiv \frac{1}{2} \sum_\alpha g^{\alpha\gamma} (g_{\alpha\beta,\mu} + g_{\alpha\mu,\beta} - g_{\beta\mu,\alpha}) \quad (2.19)$$

and the comma-derivative notation is

$$\phi_{,\mu} \equiv \frac{\partial \phi}{\partial x^\mu}, \quad (2.20)$$

such that, for example,

$$g_{\alpha\beta,\mu} = \frac{\partial g_{\alpha\beta}}{\partial x^\mu}.$$

In Equation 2.17, we explicitly reinsert the summation symbols to illustrate the number of simultaneous summations that are taking place implicitly. As one can see, this reasonably simple expression balloons quickly into something intractable by hand and time-consuming to evaluate computationally. Although the metric tensor is generally four-dimensional,³ such as the Friedmann-Lemaître-Robertson-Walker metric (Friedmann, 1922; Lemaître, 1931; Robertson, 1935; Walker, 1937) describing the expansion of the Universe,

$$g_{\mu\nu} = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & a(t)^2 & 0 & 0 \\ 0 & 0 & r^2 & 0 \\ 0 & 0 & 0 & r^2 \sin^2 \theta \end{pmatrix},$$

it often lacks a closed form, especially on large-scale chaotic behaviour that cannot be locally “washed out” (such as how on Universal scales, individual stars can be ignored to describe the homogeneity and isotropy of the Universe), and is obtained via numerical methods.

³And thus the summation indices run from 1 to 4

This is also not considering complications of writing the stress-energy tensor $T_{\mu\nu}$, which may have its own complex form (again, assuming a closed-form solution *exists*) determined via fluid-mechanical processes. All told, there are 64 equations to be solved instead of 4. In addition, the 64 solutions are “worse” to solve: harder to integrate by hand, and requiring greater time complexity. Numerical relativity codes typically simulate one star for the order of kiloseconds of its lifetime (see, for an example, ud-Doula et al., 2013), and even the most efficient codes take longer than our codes to run. For instance, the PHANTOM 3D MHD code (Price et al., 2018) can take several days to run high-resolution models. Consequently, for the number of stars we wish to simulate, we choose to adopt a Newtonian framework.

2.3 Supernovae

*And on the pedestal these words appear:
 “My name is Ozymandias, king of kings:
 Look on my works, ye Mighty, and despair!”
 Nothing beside remains.*

Ozymandias, BY PERCY BYSSHE SHELLEY

*Pallida mors æquo pulsat pede pauperum tabernas Regumque turres.
 Pale death knocks with the same tempo upon the huts of the poor and
 the towers of kings.*

 HORACE

A star’s lifetime is, unfortunately, not infinite. The key assumption underpinning stellar evolution is that of hydrostatic equilibrium: the counterplay of gravity against outwards pressure. Whilst this equilibrium holds, the star is stable and resists explosion. It is when this equilibrium is broken – with gravity the dominant force in this scenario – that the star implodes in a core-collapse supernovae (CCSN). From a thermodynamic perspective, as the star evolves, it shrinks as energy is bled off into the interstellar medium. This shrinkage compresses and heats the core, meaning the star’s specific heat capacity is negative. This system is thermodynamically unstable, and thus every star either collapses into a remnant (c.f. black holes, neutron stars) or is sustained in some form by quantum mechanical phenomena (c.f. White Dwarfs, Thorne-Żytkow objects, and other exotica).

In any case, the breaking of hydrostatic equilibrium results generally results in⁴ a supernova, which can be classified into subtypes, as shown in Figure 2.1. In this event, gravity pulls the stellar core inwards, before “bouncing” and ejecting the envelope into the interstellar medium. The physics of this approximately second-long event are highly uncertain, and are in general unable to be modelled by detailed stellar evolution codes. All models we generate – assuming they have not been terminated earlier – will terminate at the moment of supernova.

Importantly for the nucleosynthesis of heavier-than-Iron elements, the supernova shockwave propagating through the core causes a new phase of nucleosynthesis, albeit briefly.

⁴The extent of the veracity of this statement is an area of active research: Reynolds et al. (2015) presents a short recap.

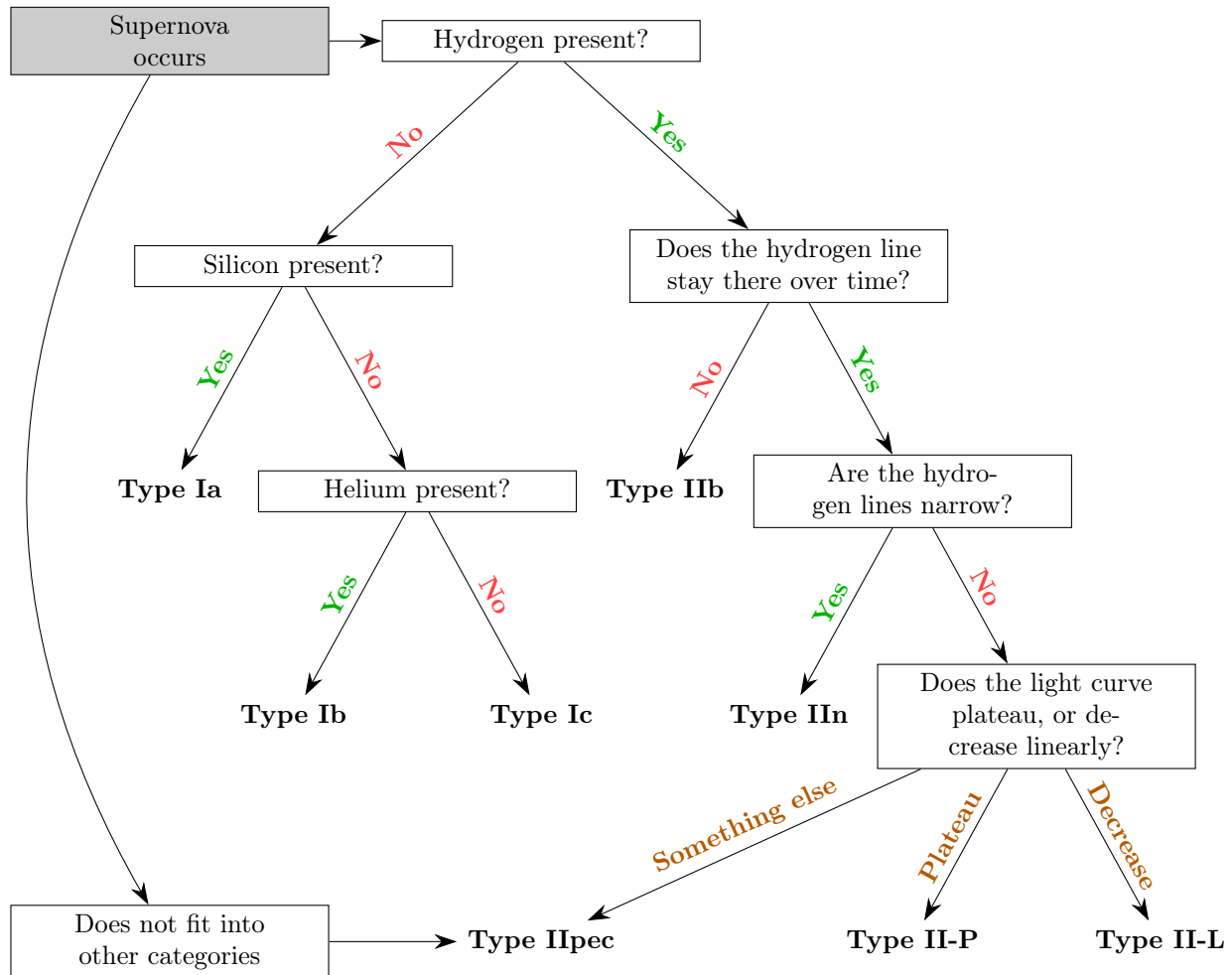


Figure 2.1: A rough outline of supernova classification. One can see that the presence of hydrogen lines distinguishes a Type I from Type II supernova. Type II supernova subtypes are determined by the behaviour of their light curves – how the apparent magnitude changes against time – and thus are determined by the luminosity of the supernova, something detailed models are not able to model. On the other hand, the presence of hydrogen/silicon/helium in the supernova is something that detailed codes *are* able to model.

Introducing a second star into the mix complicates the evolution. Whereas before a star losing mass (through, for example, a coronal mass ejection (CME)) was assumed to eject it into the ISM, it may now accrete that material directly onto its companion. In addition, either star may in principle increase in radius enough to overlap with its companion, or even touch surfaces – introducing new dynamics into the mix.

There are several formation channels for a binary star system. Binaries may be formed from birth (as discussed in section 2.1, with the caveat that in this scenario two clumps of gas form near each other), via fragmentation of a protostar’s disc, dynamical capture of one star by another (comparable to one of the suggested origins of Phobos and Deimos, per Burns (1978)), or via a higher multiplicity system ejecting one or more of its constituent stars, perhaps via a natal supernova kick.

In any case, once formed and the orbit is stabilised, binary systems obey Kepler’s three laws. In the generalised binary regime, however, the assumption that $M_1 \gg M_2$ (which underpins most statements of Kepler’s laws) is not necessarily true – for instance, the system LSS 3074 has two stellar components with ZAMS masses of approximately $30 M_\odot$ and $10 M_\odot$ (Raucq et al., 2017, and ?? of this thesis). Moreover, the fact that there are two stars in orbit rather than one object orbiting another means that Kepler’s laws must be stated more precisely. Formally, they can be expressed as (see the appendix of Richards (2021) for derivations)

1. All orbits are ellipses, with both stars orbiting a common barycentre at the focus of the ellipse,
2. A line segment joining the two stars sweeps out equal areas in equal times,
3. The square of the period of the system is proportional to the cube of the semimajor axis of the ellipse (the binary separation), with the proportionality term dependent on the sum of the masses.

The third law can be stated more precisely as,

$$T^2 = \frac{4\pi^2}{G(M_1 + M_2)} a^3, \quad (2.21)$$

and a schematic of the orbit of a binary system is given in Figure 2.2.

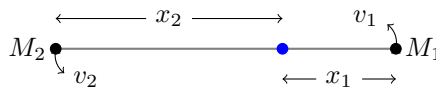


Figure 2.2: A schematic of a simple binary star system. The centre of mass (barycentre) is shown as a blue dot. Here, M_1 is the heavier star, which orbits at a smaller distance x_1 to the barycentre by Kepler’s first and Newton’s second laws. Figure adapted from Richards (2021).

When two stars are sufficiently close in proximity, events such as CMEs can cause two stars to exchange mass (whereas in the single-star case, this would be matter lost to the ISM). The nature of mass transfer is a detailed question; most binary star population synthesis codes typically inherit their mass loss prescriptions from papers such as Bondi & Hoyle (1944); Hurley et al. (2002), and this approach is what we detail in this section.

[TODO: this is a copy-n-paste from the He Stars paper – I need to rewrite it]

Our master mass-loss rate is given by

$$\begin{aligned} \dot{M} = & \dot{M}_{\text{Vink}} - 10^3 M + \frac{M(1 - dM)}{dt} + \min \left(\mathcal{R} \left(\frac{10R}{R_{\text{RL}}} \right)^3, 0.01 \right) \\ & - \min \left(\min \left[\mathcal{R} \left(\frac{10R}{R_{\text{RL}}} \right)^3, 0.01 \right], \frac{M}{T_{\text{KH}}} \mathcal{R} \left(1 - \frac{\omega}{0.8\omega_{\text{crit}}} \right) \right) \\ & - \dot{M}_{\text{Bondi}} \end{aligned} \quad (2.22)$$

where $\mathcal{R}(x) = \max(0, x)$ is the ramp function. This employs modified versions of the mass loss routines given in Vink et al. (2001), as well as Bondi & Hoyle (1944); Hurley et al. (2002). We detail the modifications we have made below. We take our accretion rates from Bondi & Hoyle (1944); Hurley et al. (2002), limiting the accretion rate by the Kelvin-Helmholtz time-scale T_{KH} of the star and by $1 - \frac{\omega}{\omega_{\text{crit}}}$. The mean accretion rate (in the absence of RLOF) onto the companion is that given in Bondi & Hoyle (1944), namely

$$\dot{M}_{\text{Bondi}} = \min \left[-\frac{1}{\sqrt{1 - e^2}} \left(\frac{GM_2}{v_{\text{W}}^2} \right)^2 \frac{\alpha_{\text{W}}}{2a^2} \frac{1}{(1 + v^2)^{3/2}} \dot{M}_{\text{Vink}}, 0.8\dot{M}_{\text{Vink}} \right], \quad (2.23)$$

where we have assumed that

$$v^2 = \frac{G(M_1 + M_2)}{av_{\text{W}}^2}, \quad (2.24)$$

and that

$$v_{\text{W}} = 2\beta_{\text{W}} \frac{GM_1}{R_1}. \quad (2.25)$$

The two free accretion parameters are assumed to be $\alpha_{\text{W}} = \frac{3}{2}$ (Boffin & Jorissen, 1988), $\beta_{\text{W}} = 7$ (Lamers et al., 1995) for O-type stars. The computation for $\dot{M}_{\text{Vink}}$ is more involved, though follows the same essential structure as Vink et al. (2001). We construct it as follows. The total pre-WR mass loss is given by de Jager et al. (1988), namely

$$\log \dot{M}_{\text{de Jager}} = - \sum_{n=0}^5 \sum_{\substack{i=0 \\ j=n-i}}^n a_{ij} \cos(i \arccos x_1) \cos(j \arccos x_2), \quad (2.26)$$

$$\dot{M}_{\text{pre-WR}} = \sqrt{\frac{Z}{Z_{\odot}}} 10^{\log \dot{M}_{\text{de Jager}}}, \quad (2.27)$$

where $x_1 = \frac{\log T - 4.05}{0.75}$, $x_2 = \frac{\log L - 4.6}{2.1}$, and the coefficients a_{ij} are given in Table VI of de Jager et al. (1988). We interpolate the bi-stability jump around 25 kK (see, for example, Pauldrach & Puls, 1990; Vink et al., 1999) with the following regime. Define the constants

$$R_1 = \left(\frac{Z}{Z_{\odot}} \right)^{0.13}, \quad (2.28)$$

$$\begin{aligned} R_2 = & -6.697 + 2.194 \log \left(\frac{L}{10^5 L_{\odot}} \right) - 1.313 \log \left(\frac{M}{30 M_{\odot}} \right) - 1.226 \log(1.3 R_1) + 0.933 \log \left(\frac{T}{4 \times 10^4} \right) \\ & - 10.92 \left(\log \left[\frac{T}{4 \times 10^4} \right] \right)^2 + 0.85 \log \left(\frac{Z}{Z_{\odot}} \right), \end{aligned} \quad (2.29)$$

and

$$\begin{aligned} R_3 = & -6.688 + 2.210 \log \left(\frac{L}{10^5 L_{\odot}} \right) - 1.339 \log \left(\frac{M}{30 M_{\odot}} \right) - 1.601 \log(0.65 R_1) + 1.07 \log \left(\frac{T}{2 \times 10^4} \right) \\ & + 0.85 \log \left(\frac{Z}{Z_{\odot}} \right). \end{aligned} \quad (2.30)$$

Our mass loss rate becomes:

$$\dot{M}_{\text{BS}} = \begin{cases} \frac{(T-T_1)10^{R_3} + (T_2-T)\dot{M}_{\text{pre-WR}}}{2.5 \times 10^3} & T_1 < T < T_2 \\ 10^{R_2} & T_2 \leq T \leq T_3 \\ \frac{(T-T_3)10^{R_2} + (T_4-T)10^{R_3}}{5 \times 10^3} & T_3 < T < T_4 \\ \frac{(6T_1-T)10^{R_2} + (T-5T_1)\dot{M}_{\text{pre-WR}}}{10^4} & 5T_1 < T < 6T_1 \\ \dot{M}_{\text{pre-WR}} & \text{otherwise} \end{cases} \quad (2.31)$$

where $T_1 = 10 \text{ kK}$, $T_2 = 12.5 \text{ kK}$, $T_3 = 22.5 \text{ kK}$, $T_4 = 27.5 \text{ kK}$. This interpolation is shown in Figure 2.3. We define

$$\mathcal{M} = \sqrt{\frac{Z}{Z_{\odot}}} \times 10^{-13.6 + 1.63 \log \left(\frac{L}{L_{\odot}} \right) + 2.22 \log X}, \quad (2.32)$$

where X is the surface hydrogen abundance of the primary, and then define

$$\dot{M}_{\text{WNL}} = \begin{cases} 5 \left(\mathcal{M} - \dot{M}_{\text{BS}} \right) (0.42 - X) + \dot{M}_{\text{BS}} & 0.42 > X > 0.4 \\ \mathcal{M} & \text{otherwise} \end{cases}. \quad (2.33)$$

Our total mass loss (provided the star has not depleted its surface helium, which will be discussed below)

is now

$$\dot{M}_A = \begin{cases} 10\dot{\mu}(\log T - 3.9) + \dot{M}_{BS} & X < 0.42 \text{ and } \log T > 3.9 \\ \dot{M}_{WNL} & X > 0.42 \text{ and } \log T > 4.0, \\ \dot{M}_{BS} & \text{otherwise} \end{cases} \quad (2.34)$$

where $\dot{\mu} = \dot{M}_{WNL} - \dot{M}_{BS}$. Finally, if the surface helium depletes we modify the mass loss rate further, defining

$$\dot{M}_{WC} = \sqrt{\frac{Z}{Z_\odot}} \times 10^{-8.3+0.84 \log\left(\frac{Z}{Z_\odot}\right)+2.04 \log X_{\text{He}}+1.04 \log(1-X_{\text{He}})} \quad (2.35)$$

such that our total mass loss rate becomes (provided $X_H < 10^{-3}$ and $\log T > 4$)

$$\dot{M}_B = \begin{cases} 100 \left(\dot{M}_{WC} - \dot{M}_{WNL} \right) (\mathcal{X} - 0.02) + \dot{M}_{WNL} & \mathcal{X} > 0.02 \\ \dot{M}_{WNL} & \mathcal{X} < 0.02 \end{cases}, \quad (2.36)$$

where $\mathcal{X} = \frac{\frac{1}{3}X_C + \frac{1}{4}X_O}{X_{\text{He}}}$. The total Vink et al. (2001) mass-loss rate is then

$$\dot{M}_{\text{Vink}} = \begin{cases} \dot{M}_B & \text{surface helium depleted} \\ \dot{M}_A & \text{otherwise} \end{cases}. \quad (2.37)$$

2.4.3

Roche Lobes

Algebra's like sheet music. The important thing isn't can you read music, it's can you hear it. Can you hear the music, Robert?

Niels Bohr, IN OPPENHEIMER

A binary system is complicated by the presence of its companions in several different ways. As alluded to in the previous section, the presence of a companion warps space-time in such a way that material can be quite distorted from a sphere or oblate spheroid but still remain gravitationally bound to the system. In the extremal case, a star can grow to fill or even exceed its Roche Lobe, which is the tear-drop equipotential shaped surface within which material remains gravitationally bound to its star. In Figure 2.4 we display these equipotential surfaces for a range of different mass ratios. It is pertinent to note that the shape of all of the Roche Lobes change rapidly at low mass ratios, however at higher mass ratios the shapes change less rapidly. In an equal-mass ratio binary, the binary configuration has rotational symmetry about the $\mathcal{L}_1$ point, and reflectional symmetry about the x -axis and the line $y = \mathcal{L}_1$.

There are two primary Roche Lobe points we consider: the $\mathcal{L}_1$ point and the $\mathcal{L}_2$ point. The $\mathcal{L}_1$ point (also known as the inner Lagrangian point) is the nexus between the two stars, the $\mathcal{L}_2$ point (the outer Lagrangian point) is located on the other side of the secondary to the inner point.

We follow the derivation given in Rose (1998) to find an expression for the Roche potential. First we note that in a rotating reference frame in Cartesian coordinates, we have

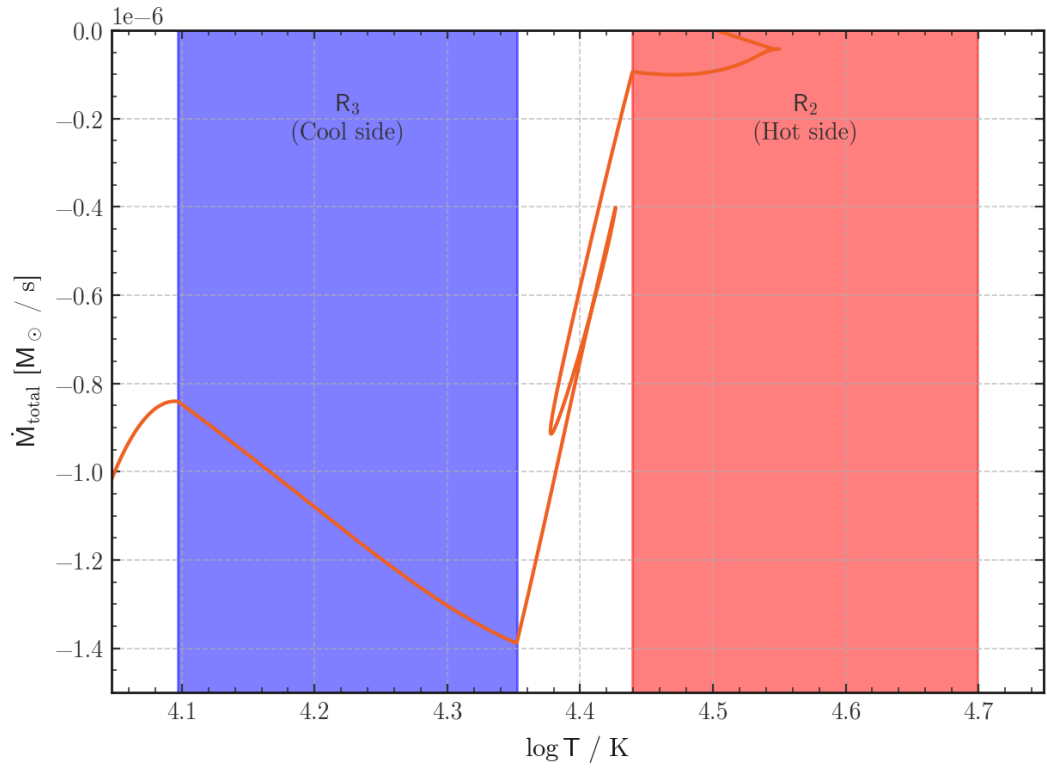


Figure 2.3: An illustration of the bi-stability jump at $\log T \approx 4.4$. This illustration was computed for a $21 M_{\odot}$ star. Shown in blue is the cool side of the bistability jump, where R_3 (Equation 2.30) is the interpolation, and shown in red is the hot side, where R_2 (Equation 2.29) is the interpolation.

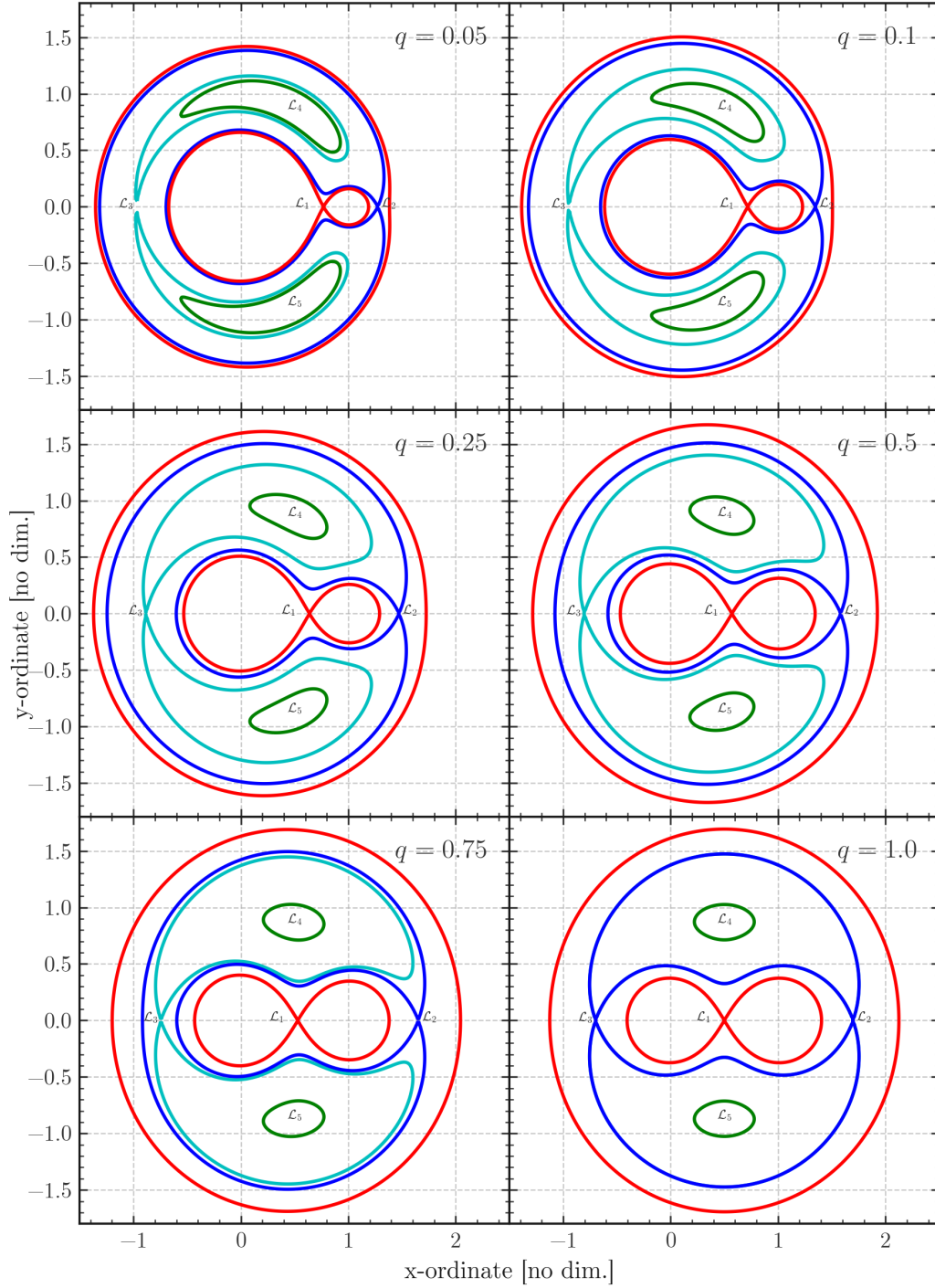


Figure 2.4: The dimensionless Roche potential as a function of mass ratio q . Note that the shape of the Roche potential changes markedly at low mass ratio, and changes more slowly as the mass ratio approaches unity. Distances have been normalised such that the primary star is at $(x, y, z) = (0, 0, 0)$ and the secondary star is at $(x, y, z) = (1, 0, 0)$, i.e., we measure distances in units of separation and that the coordinate system is centered on the primary star.

$$\overbrace{\frac{GM_1M_2}{a^2}\hat{\mathbf{x}}}^{\text{gravitational force}} = \underbrace{-M_1\vec{\omega} \times (\vec{\omega} \times \vec{r}_1)}_{\text{centrifugal force}}, \quad (2.38)$$

where $\hat{\mathbf{x}}$ is the line between the center of mass of both stars, a the separation, $\vec{\omega}$ the angular velocity, and $\vec{r}_1$ the position vector of the primary star. The x-ordinates of the centres of mass are given by

$$\left. \begin{aligned} x_1 &= \frac{M_2}{M_1+M_2}a \\ x_2 &= \frac{-M_1}{M_1+M_2}a \end{aligned} \right\} \quad (2.39)$$

Since we are considering a rotating coordinate system we have that the centrifugal force is given by

$$-M_1\vec{\omega} \times (\vec{\omega} \times \vec{r}_1) = M_1\omega^2 x_1 \quad (2.40)$$

and hence

$$\begin{aligned} \frac{GM_1M_2}{a^2}\hat{\mathbf{x}} &= M_1\omega^2 x_1 \\ \omega^2 &= \frac{GM_1M_2}{M_1x_1a^2}\hat{\mathbf{x}} \\ &= \frac{GM_2}{x_1a^2}\hat{\mathbf{x}} \\ &= \frac{GM_2(M_1+M_2)}{M_2 \cdot a \cdot a^2}\hat{\mathbf{x}} \\ &= \frac{G(M_1+M_2)}{a^3}\hat{\mathbf{x}}. \end{aligned} \quad (2.41)$$

The centrifugal potential of a test mass m at (x, y) is hence

$$V = -\frac{1}{2}m\omega^2(x^2 + y^2) = -\frac{Gm(M_1+M_2)(x^2 + y^2)}{2a^3}, \quad (2.42)$$

and thus its energy is

$$E = \frac{1}{2}m(\dot{x}^2 + \dot{y}^2) + V(x, y), \quad (2.43)$$

where

$$V(x, y) = -\overbrace{\frac{GmM_1}{\sqrt{(x-x_1)^2 + y^2}}}^{\text{G.P.E. due to primary}} - \overbrace{\frac{GmM_2}{\sqrt{(x-x_2)^2 + y^2}}}^{\text{G.P.E. due to secondary}} - \overbrace{\frac{Gm(M_1+M_2)(x^2 + y^2)}{2a^3}}^{\text{centrifugal potential}}. \quad (2.44)$$

We introduce the dimensionless coordinates measured in units of the separation of the binary (or reduced masses)

$$\xi = \frac{x}{a}, \quad \eta = \frac{y}{a}, \quad \xi_1 = \frac{M_2}{M_1 + M_2}, \quad \xi_2 = \frac{M_1}{M_1 + M_2} \quad (2.45)$$

Our potential now becomes

$$V(\xi, \eta) = -\frac{Gm(M_1 + M_2)}{a} \times \left[\frac{\xi_2}{\sqrt{(\xi - \xi_1)^2 + \eta^2}} + \frac{\xi_1}{\sqrt{(\xi - \xi_2)^2 + \eta^2}} + \frac{1}{2}(\xi^2 + \eta^2) \right]. \quad (2.46)$$

Taking the partial derivative along the ξ -axis (the rescaled x -axis), we get

$$\frac{\partial V}{\partial \xi} = -\frac{Gm(M_1 + M_2)}{a} \times \left[-\frac{\xi_2(\xi - \xi_1)}{\left((\xi - \xi_1)^2 + \eta^2\right)^{\frac{3}{2}}} - \frac{\xi_1(\xi - \xi_2)}{\left((\xi - \xi_2)^2 + \eta^2\right)^{\frac{3}{2}}} + \xi \right] \quad (2.47)$$

Hence the extrema of the potential V are at the points ξ where (taking $\eta = 0$ on the ξ -axis)

$$0 = \xi - \frac{\xi_2}{(\xi - \xi_1)^2} - \frac{\xi_1}{(\xi - \xi_2)^2}. \quad (2.48)$$

No analytical function for ξ exists unless either or both of ξ_1, ξ_2 are zero (which would only be the case for a single star), thus we must make approximations.

Roche Lobe Overflow – the $\mathcal{L}_1$ point

The inner Lagrangian point has the approximation from Eggleton (1983), which is

$$\frac{R_L}{a} \approx \frac{0.49q^{\frac{2}{3}}}{0.6q^{\frac{2}{3}} + \ln\left(1 + q^{\frac{1}{3}}\right)}. \quad (2.49)$$

We show the circle with this radius R_L in Figure 2.5 plotted against the exact solution for $\eta = \mathcal{L}_1$. Eggleton's formula is a volume-equivalent one; although his radius $R_L < \xi$, the volume of an Eggleton-Roche sphere $\frac{4}{3}\pi R_L^3$ is approximately equal (to within 1 per cent) of the numerically integrated volume, as is shown in Figure 2.6.

Roche Lobe Overflow – the $\mathcal{L}_2$ point

The outer Lagrangian point, $\mathcal{L}_2$, is given by a different zero of the potential function. The approximation we use is that of Marchant et al. (2016), where

$$R_{L2} = R_L(q) \left(1 + \frac{0.299 \arctan\left(\frac{1.84}{q^{0.397}}\right)}{q^{0.520}} \right), \quad (2.50)$$

and $R_L(q)$ is given by Equation 2.49.

We note that Marchant et al. (2016) uses an inverse mass ratio definition compared to Eggleton (1983), that is, Marchant et al. (2016) takes $q_{\text{Marchant}} = \frac{M_1}{M_2}$ whereas Eggleton (1983) takes $q_{\text{Eggleton}} = \frac{M_2}{M_1} = \frac{1}{q_{\text{Marchant}}}$. This is accounted for in Equation 2.50 such that the mass ratios are consistent (and we take $q = q_{\text{Eggleton}}$), but other texts do not necessarily differentiate or describe the difference.

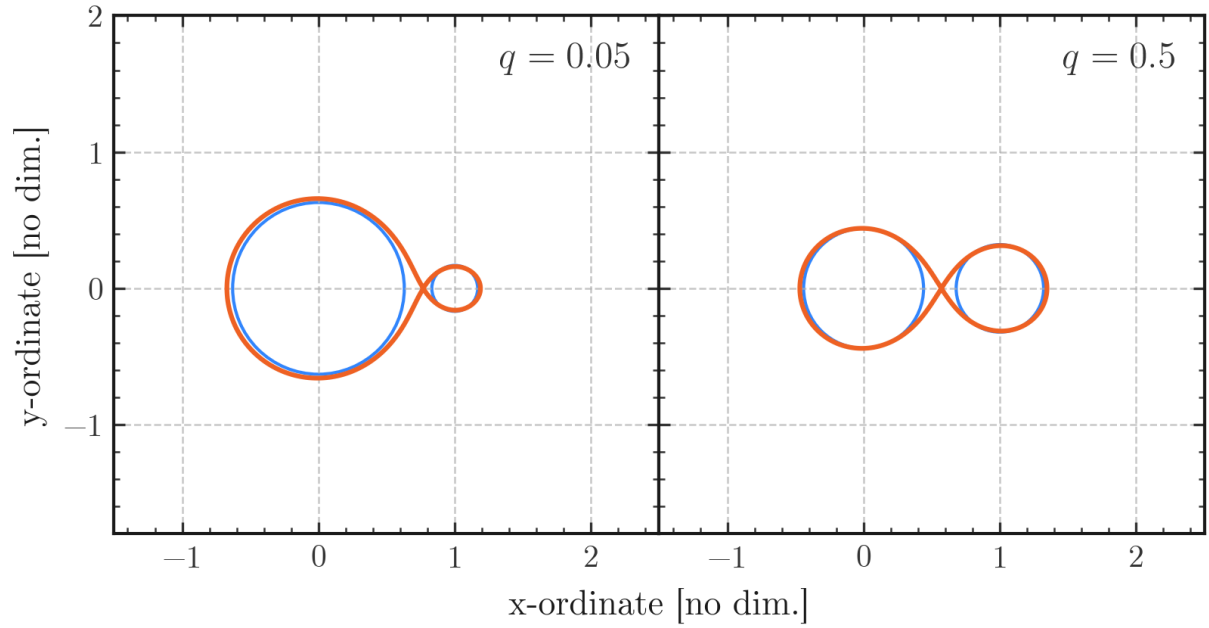


Figure 2.5: A schematic of the Roche lobe formula Equation 2.49. Here, the exact solution to the Roche lobe formula ξ in Equation 2.48 (orange) vs the fit provided by Eggleton (1983) (blue). Although precise details of the Roche lobe, such as the apex of the teardrop at the point ξ , are lost, the volume is preserved to within 1 per cent accuracy over all mass ratios.

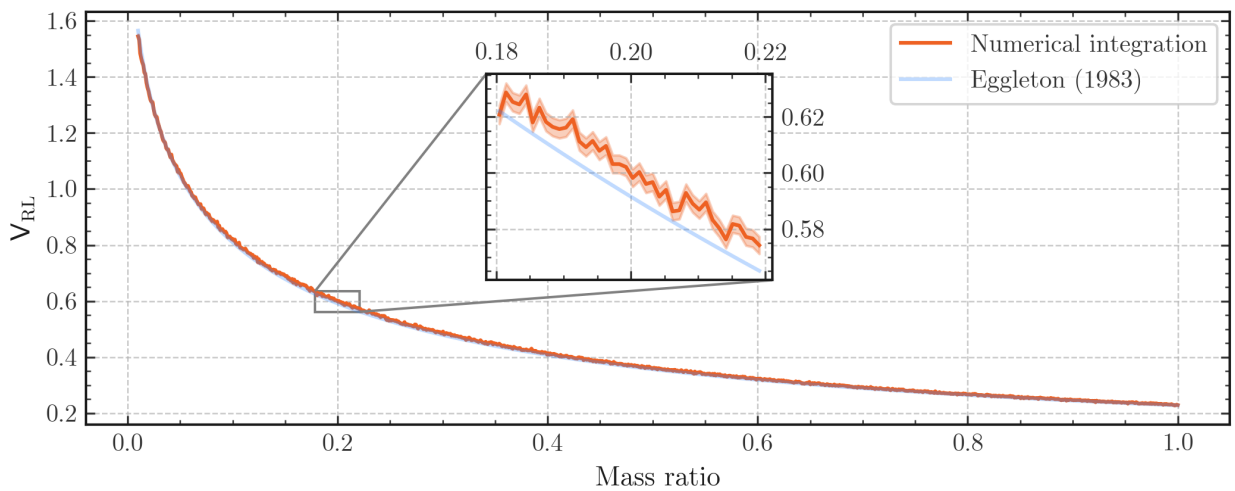


Figure 2.6: The volume of the Roche Lobe computed numerically through Monte-Carlo integration (orange) vs the volume assuming Equation 2.49 (blue). The Eggleton approximation fits well to the volume, with around 1 per cent error across the entire mass ratio range (truncated here to $q \leq 1$).

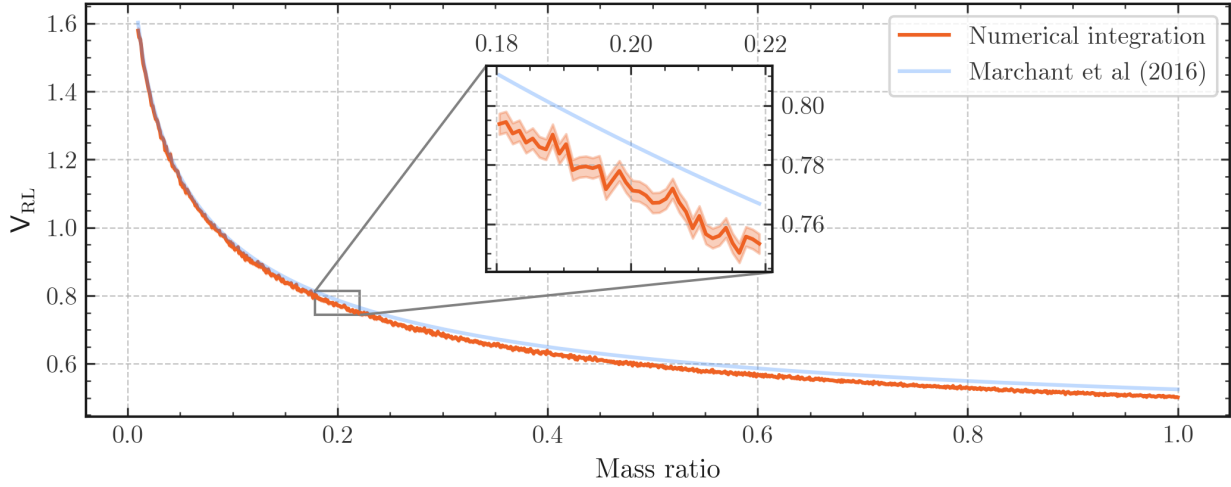


Figure 2.7: As for Figure 2.6, but for the $\mathcal{L}_2$ point with formula given by Equation 2.50 (Marchant et al., 2016).

2.5

Post-Supernova

Wit nodded.

“Oh, is this the place where I talk?” Kaladin said.

“If you wish.”

“I don’t wish. Get on with the story.”

Rhythm of War, BY BRANDON SANDERSON

After a supernova, the star leaves behind its core as a compact remnant. Generally, these are either white dwarves, black holes, or neutron stars. The interior structure of each is either complicated or, in the case of black holes, impossible to observationally probe without violating causality. Their orbital motions are now deterministic from four initial parameters: m_1, m_2, a , and e . Motion is now wholly determined by the emission of gravitational radiation through (detectable) gravitational waves. The pertinent equations of motion are given by Peters (1964), namely,

$$\frac{da}{dt} = \frac{-\beta}{a^3 (1 - e^2)^{\frac{7}{2}}} \left(1 + \frac{73}{24}e^2 + \frac{37}{96}e^4 \right) \quad (2.51)$$

$$\frac{de}{dt} = -\frac{19}{12} \frac{\beta}{a^4 (1 - e^2)^{\frac{5}{2}}} \left(e + \frac{121}{304}e^3 \right) \quad (2.52)$$

with the constant β defined as

$$\beta \equiv \frac{64}{5} \frac{G^3 m M (m + M)}{c^5}. \quad (2.53)$$

Now if one defines the constant c_0 through

$$c_0 = a_0 (1 - e_0^2) e_0^{-\frac{12}{19}} \left(1 + \frac{121}{304} e_0^2\right)^{-\frac{870}{2299}}, \quad (2.54)$$

then the time from the second supernova to the conclusion of the inspiral, termed the *coalescence time*, will be given by

$$T_c(a_0, e_0) = \frac{12}{19} \frac{c_0^4}{\beta} \int_0^{e_0} \frac{e^{\frac{22}{19}} \left(1 + \frac{121}{304} e^2\right)^{\frac{1181}{2299}}}{(1 - e^2)^{\frac{3}{2}}} de. \quad (2.55)$$

The integral has no clean closed-form solution, and a numerical integrator is typically used to compute the coalescence time. However, some approximations such as Mandel (2021) do exist and approximate the coalescence time rather well over the entire range of mass ratios.

2.5.1 Merger Rates of compact binaries

The merger rate of a population of compact binaries at time t after formation can be formalised via the heuristic

$$m(t) = \# \text{ of binaries for which } T_c + T_{\text{rej}} + T_{\text{evo}} < t, \quad (2.56)$$

where T_c is the coalescence time, T_{rej} is the rejuvenation age (the age at which the primary star last accreted mass), and T_{evo} is the evolution time-scale (the length of time the primary spends on the main sequence). $m(t)$ is called the Delay-Time Distribution (DTD) of a population.

However, the underlying assumption in this is that the population was formed after an instantaneous burst of star-forming activity at time $t = 0$, which is clearly unphysical. We must account for the so-called Star Formation Rate Density (SFRD) $\psi(z)$, assumed to be taken from Madau & Dickinson (2014):

$$\psi(z) = 0.015 \frac{(1+z)^{2.7}}{1 + \left(\frac{1+z}{2.9}\right)^{5.6}} \text{ M}_\odot \text{ yr}^{-1} \text{ Mpc}^{-3}, \quad (2.57)$$

where z is the redshift. For the purposes of converting between lookback time t and redshift z , we assume a standard LCDM cosmology with $\Omega_M = 0.3, \Omega_\Lambda = 0.7, \Omega_k = 0, H_0 = 69.6 \text{ km s}^{-1} \text{ Mpc}^{-1}$.

Of course, this ignores the effect of mixed-metallicity models, which we account for using the prescription of Langer & Norman (2006), whereupon the SFRD is multiplied by a factor χ :

$$\chi(z, Z) = \frac{\hat{\Gamma} \left[0.84, \left(\frac{Z}{Z_\odot} \right)^2 \cdot 10^{0.3z} \right]}{\Gamma(0.84)}, \quad (2.58)$$

where $\hat{\Gamma}$ and Γ are the upper incomplete and complete Gamma functions respectively. This means our Stellar Formation History (SFH) – which takes into account both redshift and metallicity – denoted $\Psi(z, Z)$, is given by

$$\Psi(z, Z) = 0.015 \int_{z_1}^{z_2} \frac{(1+z)^{2.7}}{1 + \left(\frac{1+z}{2.9}\right)^{5.6}} \frac{\hat{\Gamma} \left[0.84, \left(\frac{Z}{Z_\odot} \right)^2 \cdot 10^{0.3z} \right]}{\Gamma(0.84)} dz, \quad (2.59)$$

where z_1 and z_2 are the redshifts of our bin edges. The event rate as a function of time, $\mathcal{R}(t)$, is therefore given by

$$\mathcal{R}(t) = \Psi(z_t) * m(t), \quad (2.60)$$

where z_t is the redshift corresponding to lookback time t . Detail on how this convolution is computed may be found in Richards (2021).

2.6

A Final Note: Stellar Atmospheres

A star is not a planet – there is no firm boundary demarcating the inner of a star from the outer. The common definition of the “surface” of a star is the surface of last scattering (photosphere), which is the point radially in which the mean free path of a photon emitted from the core of the star becomes meaningfully long (internally, $\ell_\gamma \sim 2$ cm, externally, $\ell_\gamma \sim 4.4$ pc).⁵ However, outflows from the star as well as the potential interaction of a partner in a common-envelope event mean that this surface is not necessarily firm: there is no sharp turn from stellar material to empty space. Thus, the simulation of stellar atmospheres is of interest to the stellar astrophysicist. However, this introduces another rather large layer of complexity. The synthesis of a stellar spectra, for example, is not a trivial task, as well as simulating the matter content. This thesis is concerned with detailed modelling of the stellar *interior*, and thus the atmosphere is an interesting sidenote, but one we shall not pursue.

2.7

Numerical implementation

I think there is a world market for maybe five computers.

THOMAS WATSON, PRESIDENT OF IBM, 1943

In this section, we provide an overview of the four codes we either use extensively or have developed to carry out the work of this thesis. These four codes are:

1. The Cambridge STARS code (subsection 2.7.1), developed by Peter Eggleton, which we use to solve the equations of stellar structure and evolution,
2. The TUI code (subsection 2.7.2), which performs our binary population synthesis,
3. The KAITIAKI code (subsection 2.7.3), which is our grid-manager code and allows modification of the input parameters to the Cambridge STARS code programmatically, and
4. The TAKAHE code (subsection 5.1.3), which is our binary compact remnant evolution code.

⁵The mean free path is a function of the particle under consideration: for a neutrino, internally $\ell_\nu \sim 3.68$ pc and externally $\ell_\nu \sim 81$ Rpc $\sim 3 \times 10^{18}$ times the diameter of the observable Universe.

Brief History

*And through the night, behind the wheel, the mileage clicking west
I think upon Mackenzie, David Thompson and the rest
Who cracked the mountain ramparts and did show a path for me
To race the roaring Fraser to the sea.*

Northwest Passage, BY STAN ROGERS

Stellar modelling became a field of active research in the 1950s. Prior to the invention of the computer, a model was treated as a mathematical process and thus was computed – often labouriously – by hand. For instance, in Eggleton (1967), Eggleton writes that

Since we have seen in Section 3 that the structure of the flame zone can be obtained independently (almost) of the rest of the star, it is reasonable to expect that instabilities of the type considered by Schwarzschild & Härm can also be examined in this way, so that the considerable labour of computing the structure of an entire star can be avoided.

– Eggleton (1967)

The Cambridge STARS code was one of the first detailed stellar evolution codes to be written, though initially it only modelled one star in detail. To model a binary system, tweaks were made whereupon the primary would be modelled in detail whilst approximating the secondary with the equations of Hurley et al. (2002) until the primary supernova, at which point the secondary would be evolved in detail. This had its merits; notably computation time was saved, and the results were generally accurate anyway.

However, Stancliffe & Eldridge (2009) introduced detailed modelling of *both* stars in a binary system into the STARS code – that is, the interiors of both stars were computed in detail, alongside the simultaneous computation of the orbital parameters. Since then, a wide array of improvements have been made to the numerical stability of the code, as well as introducing or refining new physics.

There are four variants of this code:

1. The Cambridge STARS code, which is the version originally developed by Peter Eggleton and originally described in Eggleton (1971). This version was originally only capable of solving the equations for a single star.
2. The ELDRIDGE STARS code described in Eldridge & Tout (2004b,a); Eldridge et al. (2017), which is the version used to generate the fiducial BPASS v2 grids. This version models the primary in detail while approximating the secondary using the equations of Hurley et al. (2002).
3. The BSTARS code, which is the unmodified version lately available from Cambridge, described in Stancliffe & Eldridge (2009). This version models both stars in detail throughout their lives.
4. The AOTEAROA STARS code, which is the version of the code incorporating the changes we have made.

Unless context otherwise requires, when we refer to “the STARS code” with no further qualification, we are referring specifically to the AOTEAROA STARS code.

Matrix Inversion

The heart of STARS is a matrix inversion algorithm. Although computing the inverse of a matrix is a polynomial time process, it can still take considerable time to invert. STARS implements a Henyey matrix (Henyey et al., 1964), which is of size $n = (4 + n_{\text{comp}})N_{\text{mesh}}$, with n_{comp} the number of equations to be solved and N_{mesh} the number of meshpoints in the model. An individual element of the Henyey matrix is given by

$$\mathbf{H}_{ij} = \frac{\partial (\text{equation } i)}{\partial (\text{variable } j)}. \quad (2.61)$$

For a “typical” binary computation, with $n_{\text{comp}} = 30$ equations and $N_{\text{mesh}} = 199$ meshpoints, the matrix is already 6766×6766 , for a binary which evolves up the Asymptotic Giant Branch (AGB) (the theoretically most complex calculation we perform) with $N_{\text{mesh}} = 999$, the matrix is 33966×33966 . Introducing k new equations into the code causes the runtime of the matrix inversion calculation to scale like $\mathcal{O}(n^3 + k^3 N_{\text{mesh}}^3)$.

The matrix inversion is simple Gauss-Jordan inversion, which is of complexity class (for an $n \times n$ matrix) $\mathcal{O}(n^3)$. Coppersmith & Winograd (1987) showed that this could be optimised to $\mathcal{O}(n^{2.376})$ and conjectured reducibility to $\mathcal{O}(n^2)$.⁶ Since our code is embarassingly parallelisable, and can be executed in parallel simply by increasing the number of cores available, the payoff for decreasing the runtime by one polynomial order is inconsequential, and thus we use the straightforward implementation.

Initial Condition Selection

In this section we provide an account for the selection of our general initial conditions. Unless otherwise noted in this thesis, it is assumed that the values will be using the parameters described here and in Appendix D.

In our implementation of Mixing Length Theory (MLT) (Vitense, 1953; Böhm-Vitense, 1958; Cox & Giuli, 1968), we use an MLT parameter of $\alpha_{\text{MLT}} = 1.985$, selected via calibration against a model of the Sun. This is effectively a dissipativity parameter, and represents the depth at which a parcel of matter penetrates into a lower layer before becoming fully dissipated. The value was selected by modelling a $1 M_{\odot}$ star and varying only the mixing length parameter until it had a luminosity of $1 L_{\odot}$ and a radius of $1 R_{\odot}$, at the present age of the Sun.

Timestep control

The STARS code utilises an adaptive timestep control mechanism, whereby the timestep is allowed to shrink to model rapid changes or grow if the star is broadly inert and unchanging. The core computation is a parameter Δ , given by

$$\Delta = \frac{d_{\text{opt}}}{\sum_i \sum_j |\delta x_{ij}|} \quad (2.62)$$

where d_{opt} is a pre-defined target timestep (typically taken to be ~ 1 , unless otherwise specified; see Appendix D for our selections), and δx_{ij} is the change in variable x_i at meshpoint j . In this calculation,

⁶The current “best” optimisation is $\mathcal{O}(n^{2.3727})$ (Williams, 2012). DeepMind’s AlphaTensor has claimed a faster result in Fawzi et al. (2022), though they do not state a complexity class. This is because “...this does not yield practical algorithms.”

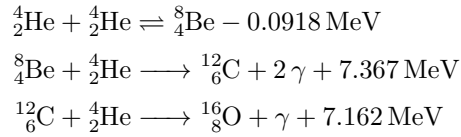
luminosity is explicitly excluded as this would cause the timestep to explode during certain points of the star's life-time. A new timestep, d' , is selected from the previous timestep d by

$$d' = \begin{cases} \min[d_1, \Delta] d & \Delta < 1 \\ d & \Delta = 1 \\ \max[d_2, \Delta] d & \Delta > 1 \end{cases} \quad (2.63)$$

d_1 and d_2 are pre-determined control parameters, most commonly we take $d_1 = 0.1$ and $d_2 = 1.2$ unless otherwise specified.

Numerical instabilities

Stellar modelling is a nontrivial endeavour and there are many cases where the code can fail seemingly arbitrarily. The largest cause of numerical instability is the helium flash, which occurs in low-mass stars between about $0.8 M_\odot \lesssim M_{\text{ZAMS}} \lesssim 2.5 M_\odot$. In this band, the triple- α process, given by the reactions (Carroll & Ostlie, 2017)⁷



occurs within a highly degenerate helium core. As a result, the temperature of the star increases sharply during this process, leading to thermal runaway. This temperature increase in turn results in an increase in the luminosity (which peaks at $\sim 1 \times 10^{10} L_\odot$). The flash itself – being the period of thermal runaway – occurs on the order of a few minutes, though the spike in luminosity lasts approximately 7 Myr (for the $2.51 M_\odot$ star shown in Figure 2.8).

A method termed “pseudoevolution” can be used to overcome this source of error. Briefly, a complete model (termed the *reference model*) of a $3 M_\odot$ star is evolved with a full model description saved at every timestep. When the helium flash occurs in a model (heuristically, $0.1 \leq M_{\text{He}}/M_\odot \leq 0.55$ and $\log T_c/\text{K} > 7.9$), we query the reference model for a model with the same helium core mass as the failed model. Using a custom subroutine in STARS, we turn off all nuclear and thermal burning and remove mass from the reference model to match the mass of the failed model. This model is then allowed to evolve as per normal, having overcome the helium flash. We note that this is not a *perfect* method, as this does induce discontinuities in the plots, as shown in Figure 2.9, though these are essentially insignificant.

The other stage where low-mass stars are prone to fail is during a thermal pulse (Schwarzschild & Härm, 1965, 1967). The mass range for thermal pulses is broader, from $\sim 0.8 M_\odot$ to $\sim 8 M_\odot$. A thermal pulse occurs as a star ascends the AGB; as a Carbon-Oxygen core (CO core) or Oxygen-Neon core (ONe core) has developed, shells of burning helium and hydrogen (separated by a region of almost pure helium) sit atop the core (Pols et al., 2001). As the intershell region dissipates due to dredge up or helium burning, hydrogen burning reignites in the hydrogen shell. As both shells burn (at the same rate), the helium-burning shell

⁷The reversible reaction is due to Beryllium-8 being unstable; its half-life is on the order of 1×10^{-16} s, and it decays back to $2 {}^4_2\text{He}$ if it does not react further.

becomes unstable and eventually a process of thermal runaway begins (similar to the helium flash, except off-core). This causes a spike in the luminosity, termed a thermal pulse. This can persist for many pulses – Stancliffe (2006) observes pulses before model termination, while Stancliffe (2005) observes 17 pulses for a model similar to those in Figure 2.8.

The method to overcome this numerical instability is involved and not fully implemented in STARS. Briefly, the number of meshpoints must be drastically increases near the onset of thermal pulses; up to 999 instead of the typical 199/499. The timestep must also be forcibly decreased; thermal pulses typically last for around a year (Pols et al., 2001) so the timestep must be reduced to not miss them (and to not have the modulus of the change in parameters – $\sum_i \sum_j |\delta x_{ij}|$ in Equation 2.62 – and thus the error, increase too much). This method was implemented to generate Figure 2.8, though only two pulses were achievable.

2.7.2

The TUI code

TUI is the core population synthesis code within the BPASS project. TUI takes as input the output from the STARS code. TUI was first described in Ghodla et al. (2023), and was more fulsomely described in Stevance et al. (2023), though we note that their version is technically a refactor of the version we use. The predominant function we use TUI for is the matching of pre- and post- supernova models for chapter 5, and in this section we provide a brief account of that functionality of TUI (though the control of the population synthesis models is beyond the scope of this thesis).

TUI has precompiled “input_” files which report a STARS plot file, an IMF weight, and a model type (merger/normal primary/normal secondary/single model/quasi-homogeneous evolution binary/single star). After loading in a model, TUI applies a given kick and computes the new orbital parameters from Tauris & Takens (1998). Crucially, a core assumption of BPASS is that the orbit remains circular after the first supernova and is allowed to be eccentric after the second. Of course, the supernova may cause the binary to become eccentric after the first supernova; TUI accounts for this by setting the separation after the first supernova to be the semi-latus rectum of the eccentric binary. TUI repeats this process after the second supernova.

For the purposes of chapter 5, if the binary remains bound after the second supernova, we record the following parameters:

1. The mass of the primary,
2. The mass of the secondary,
3. The orbital separation,
4. The eccentricity,
5. A weight term proportional to the IMF weighting,
6. The rejuvenation age, which is the time since the companion last accreted material, if it accreted more than five per cent of its initial mass (see Richards et al., 2023), and
7. The evolution age, which is the length of time from the secondary’s ZAMS until its death.

This provides us with the post-secondary supernova orbital parameters which determine the properties of the merger, such as the event rate $\mathcal{R}(z)$ and its path through period-eccentricity space.

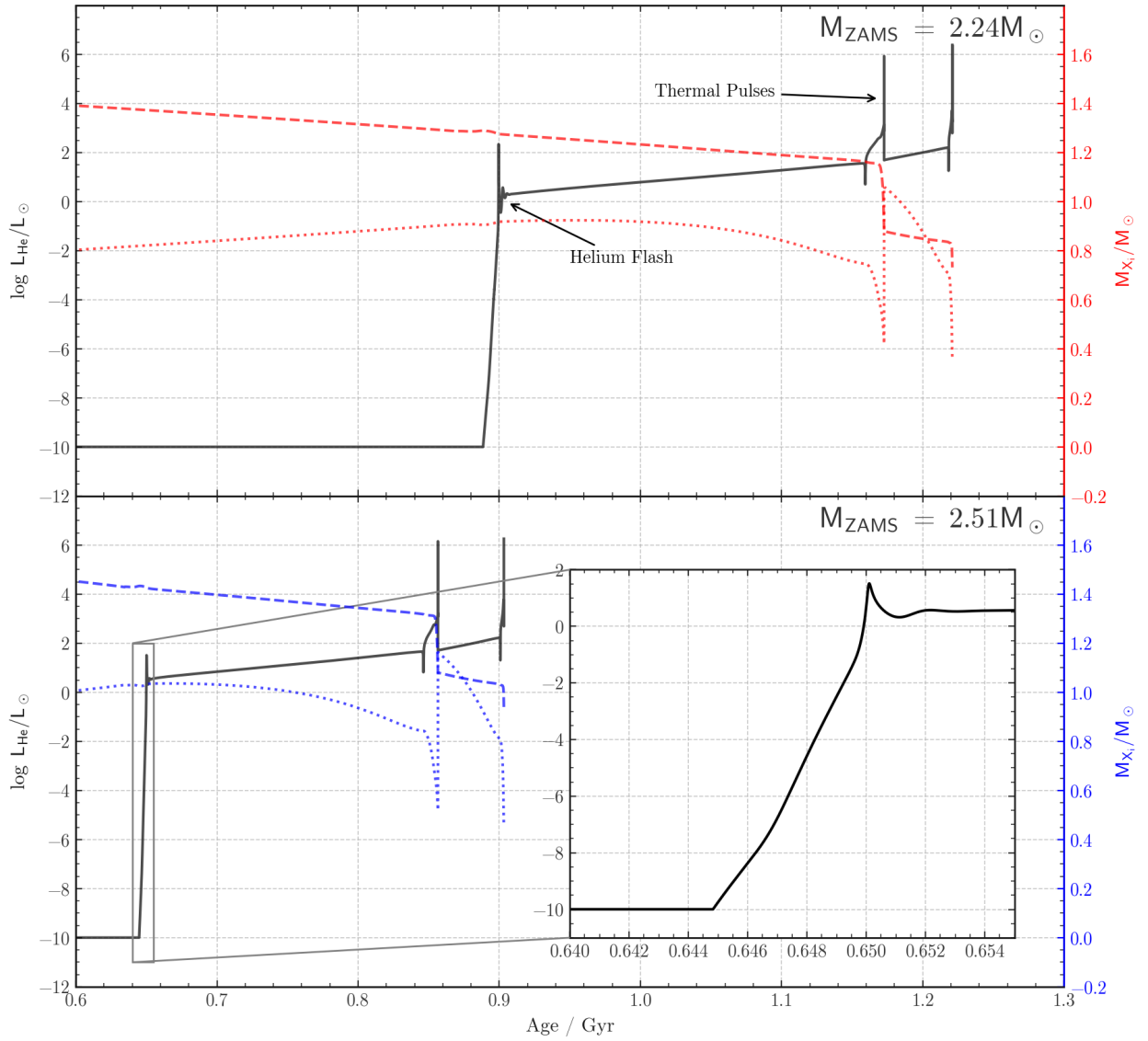


Figure 2.8: Two stars of a similar mass that undergo a helium flash and two thermal pulses. Shown in the inset is a zoomed-in helium flash that lasts ~ 7 Myr. Both stars undergo two thermal pulses before the code crashes due to the relative error in the calculation increasing too rapidly. The two stars could, in principle, undergo many more thermal pulses before their deaths.

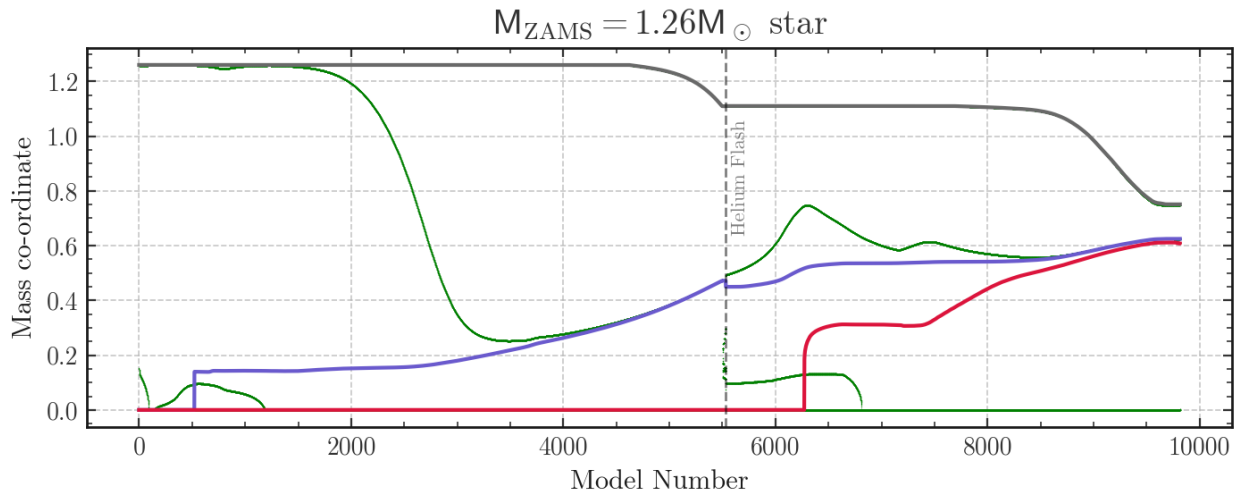


Figure 2.9: The helium flash (gray dashed line) shown on a Kippenhahn diagram. There is a slight discontinuity in the helium core mass (blue) due to a perfect match not existing in the $3 M_{\odot}$ reference model. The total mass is shown as the solid gray line, and the CO core mass is shown in red. The green lines show the boundaries of the convective regions of the star.

2.7.3 The KAITIAKI grid-watcher code

KAITIAKI is a grid-watcher code that sits atop the STARS code and manages its execution. It was developed primarily to manipulate the fixed-width text files that STARS uses, in contrast to the “typical” manipulation done through the command line.

The core of KAITIAKI is `STARSController()` interface. The most naïve run would be to initialise a model of, say, $8 M_{\odot}$ and evolve it until it evolves up the AGB. A sample of code for this is shown in Code Sample 2.1 which evolves a $10 M_{\odot}$ star.

Code Sample 2.1: A KAITIAKI script to evolve a $10 M_{\odot}$ star

```
1 import kaitiaki
2
3 # First, instantiate the STARS code. run_bs is the location of the run_bs
4 # bash script, which sets up the symlinks and actually runs the code.
5 STARS = kaitiaki.STARS.STARSController(run_bs='..')
6
7 # Loads the default data file and solar metallicity opacity table
8 STARS.blit('z020')
9
10 # Loads a 1 Msun model
11 STARS.load_default_modin()
12
13 # Sets the code up to target a ZAMS mass of 10 Msun.
```

```

14 STARS.setup_zams_inflation(10)
15
16 # Executes the code, such that the final modout will be a 10 Msun model.
17 STARS.run()
18
19 # Equivalent to "tail -399 modout > modin" -- transfers the last 2*199+1
20 # lines from modout (the last converged model) to modin
21 STARS.modout_to_modin()
22
23 # Again, to reload data
24 STARS.blit()
25
26 # Set up the key parameters for our run
27 STARS.configure_parameters({
28     'DT1': 0.1, # Fraction of timestep to allow for lower variation
29                 # (if the timestep is cut, it can be cut to 10% of its
30                 # present value)
31     'DDD': 2,   # Modulus of total desired timestep
32     'IML1': 5,  # WR mass-loss rates
33     'RML': 0,   # This was set to 10 Msun during inflation -- should be
34                 # zero so no mass is added except via the usual ML channels
35                 # (this parameter, formally, sets a constant ML rate so
36                 # that in the absence of other mechanisms, the mass would
37                 # always decrease)
38     'IMODE': 1, # Single star mode
39     'IX': 1,   # Hydrogen burning
40     'IY': 1,   # Helium burning
41     'IZ': 1,   # Metal burning
42     'ITH': 1,  # Thermal energy generation
43     'ISTART': 1 # Reset age, NMOD, and the timestep (to 10% of T_KH)
44 })
45
46 # Execute the model, run until the AGB.
47 STARS.run()

```

Looking up, he saw a thousand stars glittering in the deep velvet of a night with no moon. He knew them all, their stories and their names. He knew them in a familiar way, the way he knew his own hands.

The Name of the Wind, BY PATRICK ROTHFUSS

In this Chapter, we have explored the evolution of stars from pre-main sequence evolution to supernova, accounting for the effects of binary interaction. This represents the state of the field as at the commencement of this work. In future chapters we will detail the relevant changes we have made to these prescriptions, motivating the work we have performed.

Helium Rich Mass Transfer

*Begin at the beginning, the King said, very gravely, and go on till
you come to the end: then stop.*

Alice's Adventures in Wonderland, BY LEWIS CARROLL

As discussed in section 2.4, the presence of a companion measurably alters the evolution of a star.

3.1 Chapter Summary

Detailed Models of Overcontact Binary Stars

Post-Supernova Kinematics: The Natal Supernova Kick

*I fear not the man who has practiced 10,000 kicks once,
but I fear the man who has practiced one kick 10,000 times.*

BRUCE LEE

If you torture the data enough, they will confess.

SALVATORE VITALE

The terminus of a massive star's life is a core-collapse supernova, where the hydrostatic equilibrium between the pressure of the star and its self-gravity is broken, and the star collapses in on itself. Once the star reaches its minimum radius, it rebounds, ejecting an amount of mass we denote M_{ej} from the star and leaving a remnant of mass M_{rem} behind. If the ejection is asymmetric, the collapse endows the nascent remnant with a velocity v_{k} whose overall direction is antialigned to the more massive ejecta. This motion is termed a natal supernova kick.

Given an observation of the proper velocity of a compact remnant, it is in principle possible to trace its motion back and ascertain its kick velocity. However, this is not necessarily a straightforward endeavour. If the proper velocity is high and the remnant old, it has travelled quite a distance from its explosion site to be located at the point the observation has been taken. In addition, the star has potentially had its proper velocity modified by gravitational attraction from either other runaway stars or its host galaxy's potential. One would require the exact location in 3D space as well as the 3D velocity and remnant age in order to ascertain the motion, but from there the inverse problem becomes straightforward.

5.1

Overview

As a consequence of conservation of momentum, it can be shown that if the ejecta is assumed to eject along a preferred axis, the velocity of the kick can be written as

$$v_{\text{k}} = \alpha \left(\frac{M_{\text{ej}}}{M_{\text{rem}}} \right) + \beta \left(\frac{M_{\text{NS}}}{M_{\text{rem}}} \right), \quad (5.1)$$

for free parameters α and β (both measured in km s^{-1}). Here, M_{NS} is the mass of a Neutron Star (NS); if the kick is applied to a NS, then $M_{\text{NS}} = M_{\text{rem}}$ and the coefficient reduces to unity. If the kick is applied to a Black Hole (BH), then we can take $M_{\text{NS}} = 1.4 M_{\odot}$, which serves to convert the distribution from one in velocity-space to one in momentum-space; see Bray & Eldridge (2016, 2018); Richards et al. (2023) for further detail.

This parameterization introduces, effectively, four free parameters in $\alpha, \beta, \theta, \phi$ (the latter two being azimuthal and equatorial angles of the ejecta mass). We assume that the preferential axis is sampled uniformly

across the 4π steradians of the star, and that our only two free parameters are α and β . It is pertinent to note two things due to this assumption: firstly, we are not assuming that the ejection occurs along one axis and that the kick is aligned along a different, random, axis. Rather, we are assuming that the asymmetric ejection occurs along a randomly sampled axis and that the kick is antialigned to that. The second note, explored further in subsection 5.1.2, is that there is considerable uncertainty around the nature of the kick and the precise mechanism that causes it to eject along a given axis. This is the reason we sample the kick direction uniformly, though we will simulate sufficiently many systems (on the order of $\mathcal{O}(10^{12})$) that this will be marginalised over.

The existence of two free parameters, α and β , mean that a choice on the values must be made. The “traditional” way to choose these parameters, as outlined in Bray & Eldridge (2016, 2018), is to refine them based on isolated pulsar velocities. This methodology, however, has significant drawbacks. Notably, the reliance on one observation can engender biases in the result at the expense of no longer predicting accurate results for other observables. For instance, in the limit of ultra-stripped supernovae (USSNe) where the ejecta-remnant mass ratio becomes small, the fiducial kick in Bray & Eldridge (2018) would produce kick velocities that are strongly negative (indicating a strong kick aligned with the direction of motion of the ejecta), contrasted with the weak kicks we expect from these supernovae (e.g., Willcox et al., 2021). It becomes evident that a new pipeline is needed, whereupon more constraints than simply pulsar velocities are tested. In this Chapter, we outline four such tests that form the basis of the work of Richards et al. (2023). These tests – refining on observed LVK Collaboration merger rates, period-eccentricity distributions, isolated pulsar velocities, and production of USSNe – allow us to narrow the parameter space significantly, with the promise of future refinement with more observations (such as through O4).

5.1.1 Parameter space

We perform our computation on three populations, termed FINEGRID, WIDEGRID, and HOBBS, whose properties are outlined in Table 5.1. These populations simulate 1,000 kicks per system sampled. The aim of the three populations is twofold. Firstly, by generating FINEGRID and WIDEGRID, we are able to look both at fine detail in the region at which the most likely value is expected to lie and also the large-scale behaviour over wider parameter ranges.

Population Name	α_{low} [km s ⁻¹]	α_{high} [km s ⁻¹]	β_{low} [km s ⁻¹]	β_{high} [km s ⁻¹]	Step Size [km s ⁻¹]
FINEGRID	0	400	-200	200	10
WIDEGRID	0	700	-350	350	50
HOBBS	-	-	-	-	-

Table 5.1: The populations under study. HOBBS is a fiducial kick prescription as given by Hobbs et al. (2005), which is a Maxwellian distribution with distribution parameter $\sigma = 265 \text{ km s}^{-1}$, based on an empirical fit to the velocities of 233 pulsars. All three prescriptions simulated 1,000 kicks for each system, with the resultant kick being the mean velocity sampled, and the direction of the kick being sampled uniformly across the 4π steradians of the orbit.

Secondly, by using Hobbs et al. (2005), we are able to create a population governed by a kick that has been well-tested in literature (see, for example, Bouffanais et al., 2021; Broekgaarden et al., 2021; Fragione,

2021; Igoshev et al., 2021; Mandel & Broekgaarden, 2022) which serves as a fiducial comparison. From a Bayesian perspective, we are essentially choosing the HOBBS distribution as our prior for the express purpose of a Bayesian test in section 5.3.

The choice of extremal values for α and β are somewhat arbitrary. Figure 5.1 shows a wider parameter space with $\alpha \in [0, 1000], \beta \in [-500, 500]$. Regions in the upper right (bottom left) corner of the plot have quite high (low) mean kick velocities. Although both grids encroach on these regions somewhat, there is a tradeoff between exploring the grid space fully and using computational resources efficiently. Our mean kick velocities should not be too extreme; the mean for a Hobbs et al. (2005) kick is $\sigma = 265 \text{ km s}^{-1}$, for a Verbunt et al. (2017) is $\sigma_1 = 75 \text{ km s}^{-1}$ and $\sigma_2 = 316 \text{ km s}^{-1}$, and for the observed Willcox et al. (2021) kick $\sigma_1 \approx 30 \text{ km s}^{-1}, \sigma_2 \approx 230 \text{ km s}^{-1}$. These¹ are shown on the grid as black lines.

Whilst the exact parameters can be shifted somewhat and maintain the statistical integrity given these constraints, there are a few additional heuristic constraints we place on the regions to select. First, we wish to capture low-velocity USSN kicks (as will be discussed in section 5.5), necessitating a region near $\alpha = \beta = 0$ (where the kick would be identically zero for all ejecta-remnant mass ratios). Secondly, the fiducial work of Bray & Eldridge (2018) found $\alpha = 100, \beta = -170$. Although a goal was to (ideally) remove $\beta < 0$ as a possible value (for USSN again), to impose that as a prior when the fiducial value sat within that range would have gone contrary to typical Bayesian practices. Thirdly, we expected but could not guarantee that the magnitude of each parameter would shrink with further tests, given the large uncertainty bands that Bray & Eldridge (2018) found (hence broadening our search area beyond $|\alpha| = 100, |\beta| = 170$). Finally, we elected to increase the grid search region by 75 per cent for WIDEGRID as this would allow us to just reach $\bar{v}_k = 700 \text{ km s}^{-1}$, which is where the Cumulative Distribution Function (CDF) of the Hobbs et al. (2005); Verbunt et al. (2017); Willcox et al. (2021) distributions all (approximately) begin to level out (signifying few velocities above those), as shown in Figure 5.5.

5.1.2

Natal Kick Prescriptions

There are conflicting accounts as to the cause of the kick (see, for example, Fryer et al., 1998; Arzoumanian et al., 2002; Bombaci & Popov, 2004; Hobbs et al., 2005; Bray & Eldridge, 2016; Verbunt et al., 2017). There are conflicting accounts as to the form and distribution of the kick too, as discussed in, for example, Blaauw (1961); Gunn & Ostriker (1970); de Loore et al. (1975); Sutantyo (1978); Hobbs et al. (2005); Baker et al. (2008); Wongwathanarat et al. (2013); Bray & Eldridge (2018); Giacobbo & Mapelli (2020); Mandel & Müller (2020).

We stress that we are not attempting to prescribe *which* prescription is somehow “best”, but seek an easy-to-configure kick which allows us to configure kicks without changing any of the physics of how our population was formed. This can be done by modifying the kick of Bray & Eldridge (2016), which takes two free parameters which can be changed without imposing constraints on physics. This prescription is hereinafter referred to as the “Bray kick”. Similarly, we require for the purposes of computing log-Likelihoods in section 5.3 a comparison kick, for which we select the kick of Hobbs et al. (2005), a well-known empirically derived kick. This prescription is hereinafter referred to as the “Hobbs kick”.

The Bray kick velocity is prescribed by Equation 5.1. The coefficient to the β term, $\frac{M_{\text{NS}}}{M_{\text{rem}}}$, serves to

¹Due to the grid size and the random sampling, a true contour is difficult to observe due to sampling noise. The contours shown are thus linear fits to the actual contours.

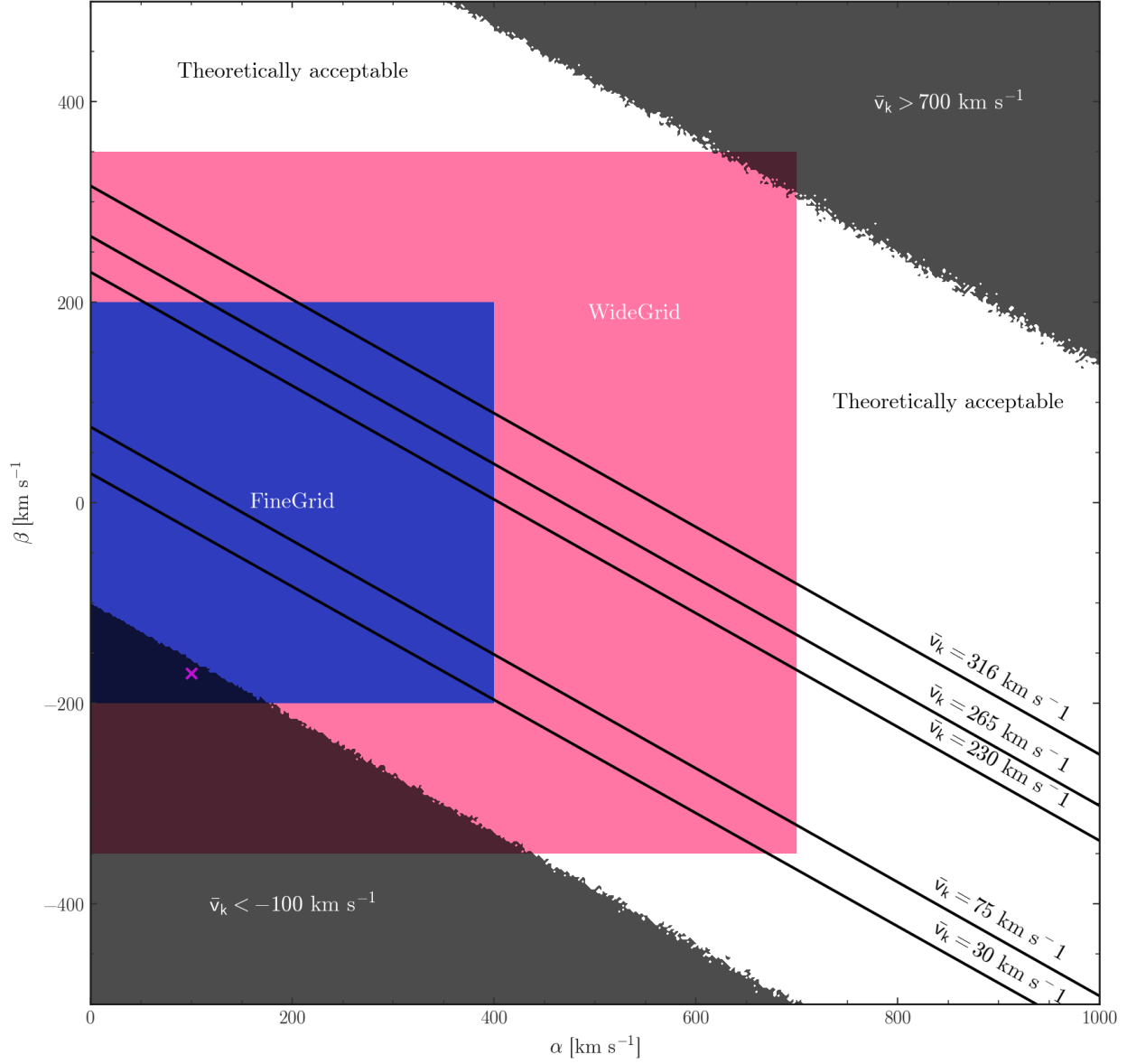


Figure 5.1: A schematic of how FINEGRID and WIDEGRID are related. The mean kick velocity v_k was computed by averaging the kick velocities of 1,000 simulated kicks for each combination of α and β , with $m_{\text{rem}} \in [0.1, 2.5]$ and $m_{\text{ej}} \in [0.1, m_{\text{rem}}]$ sampled uniformly. The pink cross represents the point $\alpha = 100, \beta = -170$, which are the fiducial kick parameters from Bray & Eldridge (2018).

generalise the kick from purely NSs to BHs as well. When applied to a BH, M_{NS} is taken to be $1.4 M_{\odot}$, and conversely when applied to a NS, M_{NS} is by definition the remnant mass, so the kick velocity becomes

$$v_k = \alpha \left(\frac{M_{\text{ej}}}{M_{\text{rem}}} \right) + \beta. \quad (5.2)$$

On the other hand, the Hobbs kick velocity is given by the Probability Density Function (PDF)

$$P(v_k) dv_k = \sqrt{\frac{2}{\pi}} \frac{v_k^2}{\sigma^3} \exp \left(-\frac{v_k^2}{2\sigma^2} \right) dv_k, \quad (5.3)$$

where $\sigma = 265 \text{ km s}^{-1}$ is the distribution parameter found by fit in Hobbs et al. (2005) to the velocities of 233 pulsars.

5.1.3 The TAKAHE binary evolution code

As part of this work, we have developed the TAKAHE binary evolution code, which solves the coupled equations of motion Equation 2.51 and Equation 2.52. Originally, to facilitate the equations being solved rapidly, we wrote the code as a PYTHON wrapper to a JULIA (Bezanson et al., 2017) integrator. However, the PYJULIA library (Arakaki et al., 2023) proved to be fragile² and is not in fact supported officially on recent Python versions. To facilitate the potential interoperability between KAITIAKI and TAKAHE (and to remove a prolonged loading time when the JULIA runtime was loaded, given that PYJULIA precompiles our modules), we elected to rewrite the core integrator into PYTHON and to optimise it using traditional PYTHON methods instead.

There are two relevant functions in TAKAHE. The remaining functions are utility functions that allow us to (for instance) convert between lookback time / redshift, and period / separation. These functions are `integrate_eoms()` (Code Sample A.1), which solves the coupled differential equations for $\frac{da}{dt}$ and $\frac{de}{dt}$ (Equation 2.51 and Equation 2.52), and `integrate_timescale()` (Code Sample A.2), which solves Equation 2.55 for the coalescence time of the binary. With sufficient performance optimisations,³ `integrate_timescale()` is able to achieve speeds of approximately 18,000 numerical integrations per second on an office PC⁴ on a sample of Neutron Star - Black Hole Binary (NS-BH) binaries in the LISA dataset (Tang, 2024).

Algorithmically, both codes employ a simple Euler integration routine (where $f_{n+1} = f_n + \frac{df_n}{dt} dt$) and the choices work as follows:

1. When `integrate_eoms()` takes a step in time, it evaluates da and de . It then computes the relative changes in both e and a that arise by taking this timestep (through $\delta f = \left| 0.01 \frac{f}{f} \right|$), and takes a timestep corresponding to the smallest change between a and e , or half of the width of the smallest BPASS time bin.
2. `integrate_timescale()` breaks up Equation 2.55 into four cases dependent on the initial eccentricity. Let τ be the coalescence time of a circular binary ($\tau = \frac{a_0^4}{4\beta}$), then the cases are:

²It requires custom installation procedures (so cannot be included as a dependency in e.g. a PYTHON package), would hang with no discernible output or ability to ascertain why, and could not handle KeyboardInterrupt exceptions in PYTHON

³Primarily, just-in-time compilation with the NUMBA (Lam et al., 2015) library.

⁴A Dell OptiPlex 7480 AIO with a 12-core 3.1 GHz Intel Core i5-10500 CPU, running no other programs except for during the final few iterations

$$t_c = \begin{cases} \tau & e_0 = 0 \\ \frac{c_0^4}{4\beta} e_0^{\frac{48}{19}} & e_0 < 0.01 \\ \frac{12}{19} \frac{c_0^4}{\beta} \int_0^{e_0} \frac{e^{\frac{22}{19}} \left(1 + \frac{121}{304} e^2\right)^{\frac{1181}{2299}}}{(1-e^2)^{\frac{3}{2}}} de & 0.01 \leq e_0 \leq 0.99 \\ \tau \frac{768}{425} (1 - e_0^2)^{\frac{7}{2}} & 0.99 < e_0 \end{cases}, \quad (5.4)$$

where all symbols have the same meaning as in section 2.5. The integral in the intermediate case is evaluated numerically, taking a fixed timestep of $de = \frac{e_0}{1000}$ such that the discretised sum becomes nothing more than

$$t_c (0.01 \leq e_0 \leq 0.99) = \frac{e_0}{1000} \frac{12}{19} \frac{c_0^4}{\beta} \sum_{e=0}^{e_0} \frac{e^{\frac{22}{19}} \left(1 + \frac{121}{304} e^2\right)^{\frac{1181}{2299}}}{(1 - e^2)^{\frac{3}{2}}}, \quad (5.5)$$

and dependence of the coalescence time t_c on the initial separation a_0 is handled through the c_0 term. Since the step size is an arbitrary choice, a tradeoff can be made between accuracy and runtime. Figure 5.2 shows the effect of changing the step size ($\frac{e_0}{1000}$ in Equation 5.5) on convergence to a final result, and Figure 5.3 shows the runtime as a function of step size on an office computer.

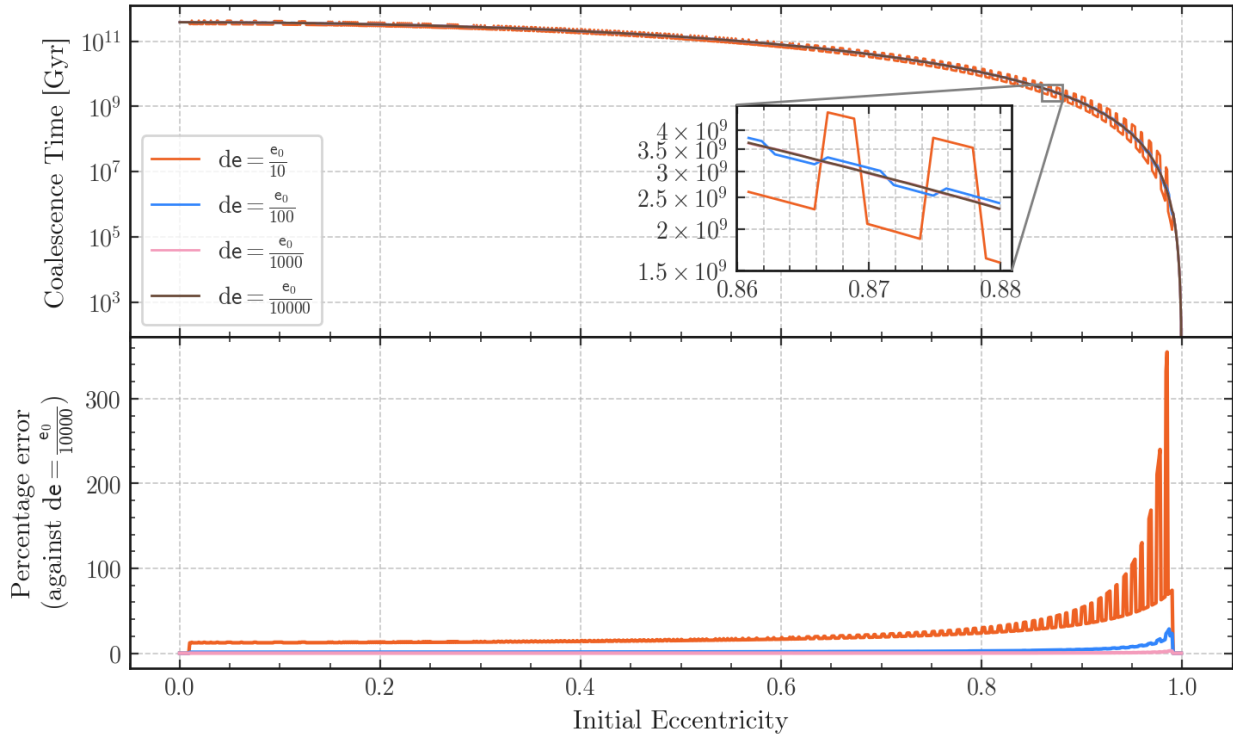


Figure 5.2: The effect of varying de in Equation 5.5 on the coalescence time. Although hard to see, the pink line does oscillate above and below the brown line. This binary was assumed to have $m_1 = m_2 = 1.44 M_\odot$ and $a_0 = 1970.5 R_\odot$ (the mean separation of a sample of binaries with $\alpha = 100 \text{ km s}^{-1}$, $\beta = -170 \text{ km s}^{-1}$).

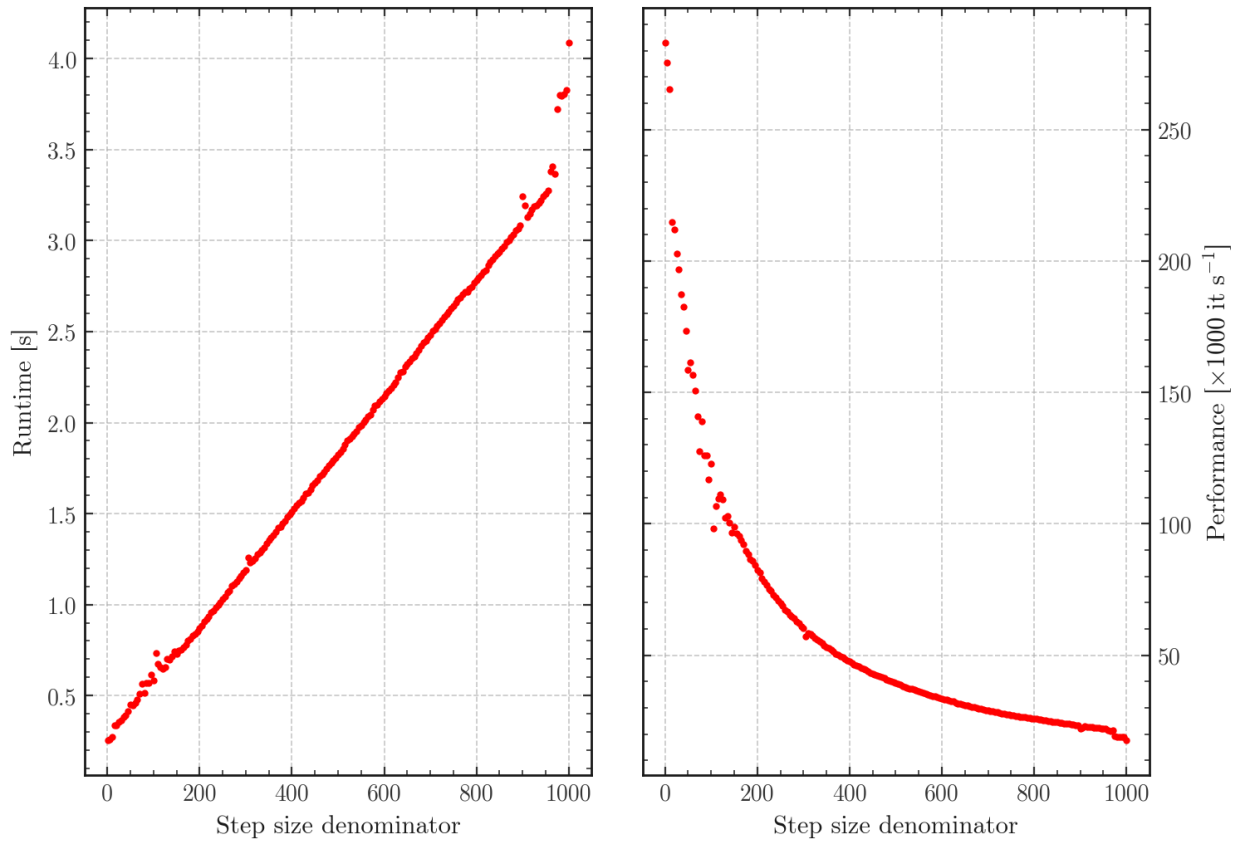


Figure 5.3: The effect of varying d_e in Equation 5.5 on the runtime, which is demonstrably $\mathcal{O}(n)$, and the performance of the algorithm. The input data to this algorithm were the LISA binary sample from Tang (2024). The performance is shown in units of thousands of iterations per second.

Merger Rates of BNS systems

The first test is to match the observed merger rates of Binary Neutron Star (BNS) systems. To do this, we use the LVK Collaboration catalogue of results from Operating Run 3 (O3). The simplest test is to match the merger rate today, which we denote as $\mathcal{R}_0$, to that observed by LVK Collaboration (LIGO-Virgo, 2021; LIGO-Virgo-KAGRA, 2021). We display this in Figure 5.4 alongside the previously reported fiducial values of $\alpha = 100 \text{ km s}^{-1}$ and $\beta = -170 \text{ km s}^{-1}$, the theoretical merger rate calculation performed by Eldridge et al. (2019), and the observed merger rates from LIGO-Virgo (2021) and LIGO-Virgo-KAGRA (2021). We term the contour defined by LIGO-Virgo (2021) (in black) the “LIGO contour”.

Selecting the value to calibrate to is not a straightforward task. Although the LIGO-Virgo (2021) contour is superseded by the LIGO-Virgo-KAGRA (2021) contour, the separation into different inference models means that the merger rates for BNS systems have intractably large error bars covering the entire plane; for instance, their MULTISOURCE (MS) model alone spans $2.11 \leq \log \mathcal{R}_0 \leq 3.23$ which covers FINEGRID entirely and almost entirely covers WIDEGRID. In addition to the lack of constraint, we would be imposing additional parameters into the model insofar as the selection of model – whether, for example, PDB was taken as truth over BGP, MS, or one of the other population models. Whilst an interesting and assuredly insightful endeavour, the aim of this work is instead to develop a simple test that can be used in future on extra populations such as these. A simple criterion which can be extended is valuable, and future work may extend these tests in multiple directions.

Period-Eccentricity Distributions

Since in the absence of any mass transfer the motion of two compact binaries is wholly dependent on their initial separation and orbital eccentricity, they trace out paths in a two-dimensional phase space. In general, since binaries circularise over time, and by the emission of gravitational waves their orbits shrink, a binary will start in the upper-right region of period-eccentricity space, and evolve towards the bottom left. As a result, given a population of binary stars, we can design a density plot, where the density is effectively the likelihood of a binary selected at random having the parameters we sample.

Log-Likelihood calculations

Computationally, for each binary in our sample, we trace its path through period-eccentricity space, and at every timestep we add $w \times dt$ to the density plot, where dt is the timestep, and w is the weight term output from the TUI code proportional to the IMF weight (where the constant of proportionality is related to the uniformly sampled kick direction). The result of this computation, for the parameter values that lie along the LIGO contour, are shown in Appendix C. Given a catalogue of N (observed) binary stars, we can then compute the likelihood of the model as

$$\ln \mathcal{L} = \sum_{n=0}^N \ln p(\ln P_n, e_n), \quad (5.6)$$

where $p(\ln P_n, e_n)$ is the content of the density plot at the location of the n^{th} observed BNS system.

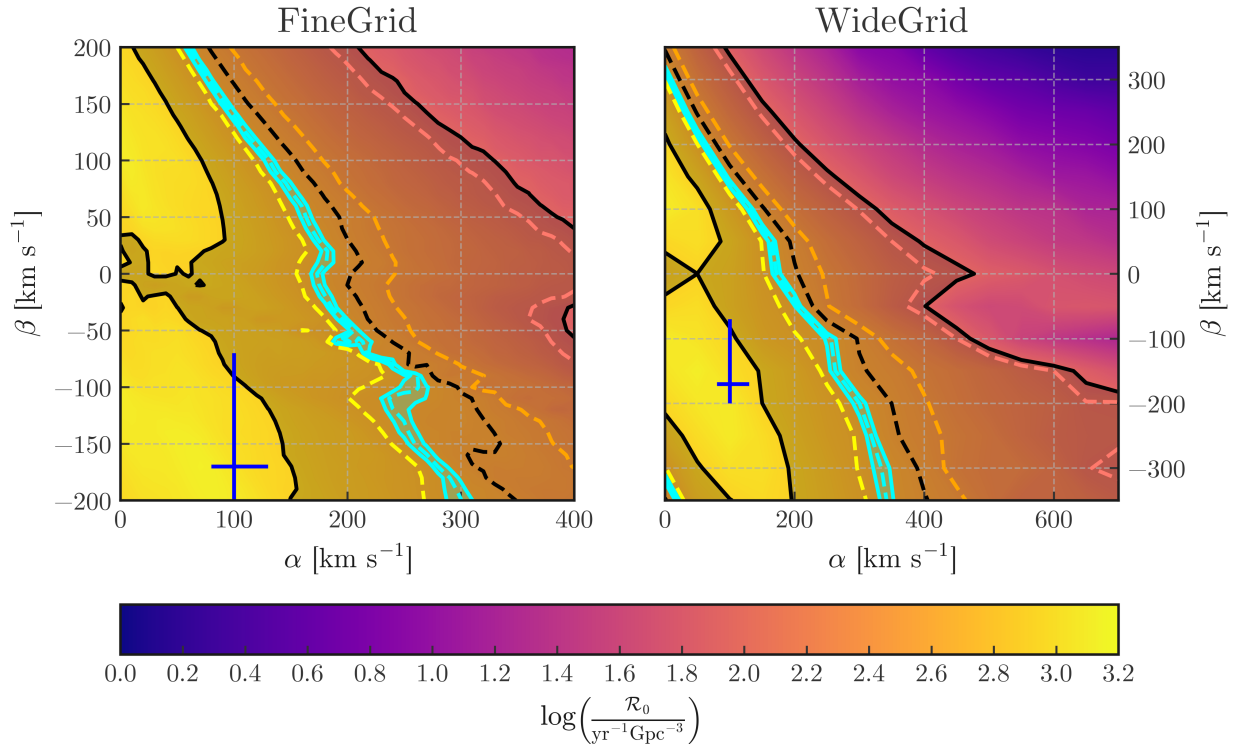


Figure 5.4: The merger rate as a function of kick parameters α and β . The cyan contour is the theoretical merger rate as reported by Eldridge et al. (2019). The black contour is the merger rate observed in LIGO-Virgo (2021). The remaining contours are the merger rates reported in LIGO-Virgo-KAGRA (2021) (pink: BGP, yellow: MS, orange: PDB [ind], see LIGO-Virgo-KAGRA (2021) for an explanation of each model). Shaded regions represent uncertainties, and uncertainties are not reported for the LIGO-Virgo-KAGRA (2021) results as they would cover the entire plane. The blue cross is the fiducial value of α and β from Bray & Eldridge (2018). This value lies off the previous work’s contours as the rejuvenation age calculation has been improved in TUI in preparation for the next major BPASS release.

Name	$\log(P/\text{days})$	e	Original Reference	Name	$\log(P/\text{days})$	e	Original Reference
J0453+1559	0.609808	0.113	Martinez et al. (2015)	J0737-3039	-0.991400	0.088	Kramer et al. (2006)
B1534+12	-0.375718	0.274	Fonseca et al. (2014)	J1756-2251	-0.494850	0.181	Faulkner et al. (2005)
B1913+16	-0.490797	0.617	Hulse & Taylor (1975)	J1913+1102	-0.686133	0.090	Lazarus et al. (2016)
J1757-1854	-0.735182	0.606	Cameron et al. (2018)	J1518+4904	0.936212	0.249	Janssen et al. (2008)
J1811-1736	1.273672	0.828	Corongiu et al. (2007)	J1829+2456	0.070407	0.139	Champion et al. (2004)
J1930-1852	1.653791	0.399	Swiggum et al. (2015)	J1753-2240	1.134751	0.304	Keith et al. (2009)
J1411+2551	0.417638	0.169	Martinez et al. (2017)	J1946+2052	-1.107905	0.064	Stovall et al. (2018)

Table 5.2: The systems we use in our study. These data were originally compiled in Vigna-Gómez et al. (2018).

Then, the Bayes factor (defined as the likelihood ratio) can be computed as

$$\mathcal{K} \equiv \frac{\mathcal{L}}{\mathcal{L}_{\text{Hobbs}}} \implies \ln \mathcal{K} = \ln \mathcal{L} - \ln \mathcal{L}_{\text{Hobbs}}, \quad (5.7)$$

where $\mathcal{L}_{\text{Hobbs}}$ is the log-Likelihood of the Hobbs kick. We use a catalogue of fourteen BNS systems originally compiled in Vigna-Gómez et al. (2018), shown in Table 5.2. From this catalogue, we sample the period-eccentricity distribution at that space and take that as the probability of observing that binary, thus allowing us to use the content of the period-eccentricity distribution as the $\ln p$ term in our summand of Equation 5.6.

5.4 Single-Star Kick Velocities

The traditional methodology for refining a kick prescription is to consider the velocity distribution of isolated pulsars. Thus, given a suitable observational dataset, we would be able to compute the expected kick distribution and compare it, via a Kolmogorov-Smirnov (K-S) test, to the observed data set. The K-S statistic between distributions $\mathcal{F}$ (the observed distribution) and $\mathcal{G}$ (the compared distribution) is given by

$$\text{KS} = \max |\text{cdf}(\mathcal{F}) - \text{cdf}(\mathcal{G})|, \quad (5.8)$$

where cdf means the CDF function, defined as the integral of the PDF. Given the CDF is bounded between 0 and 1 inclusive, the K-S statistic is also bounded between this region. A K-S statistic of 1 means that the compared distribution diverges perfectly from the observed distribution, whereas a K-S statistic of 0 means it matches perfectly. This test of divergence was the refinement performed in Bray & Eldridge (2016, 2018), hence we select this (with an independently generated dataset to those works) to be consistent with prior work.

5.4.1 Pulsar dataset

We use the same dataset of 81 pulsars from Willcox et al. (2021), and filter the dataset in the same way that they do. We remove millisecond pulsars, pulsars in globular clusters, and pulsars known to be in binaries,

in order to homogenise the dataset. The PDF and CDF of Willcox et al. (2021), as well as those generated from Hobbs et al. (2005) and Bray & Eldridge (2018), is displayed in Figure 5.5.

This homogenised dataset gives the transverse velocities of each pulsar alongside the RA/Dec, the proper motion in the RA/Dec directions, and parallax. We assume that the transverse velocity is approximately the original kick velocity. As mentioned earlier in this chapter, old remnants with high velocities may have moved substantially – a 20 Myr remnant with a transverse velocity of 500 km s^{-1} will have moved $\sim 10 \text{ kpc}$, absent external forces (Willcox et al., 2021). However, this is likely only to be an issue for pulsars whose transverse velocity substantially exceeds the transverse Galactic rotational velocity of the Sun $v_{\text{TGRV}} \approx 230 \text{ km s}^{-1}$ (Bhattacharjee et al., 2014), shown in yellow on Figure 5.5. Although the dataset of Willcox et al. (2021) has pulsars with velocities higher than this value – peaking at $\sim 850 \text{ km s}^{-1}$ – this is not a grave concern for this work, for two reasons. Firstly, the kick velocities output from TUI are taken immediately after the supernova. Secondly, the analysis Willcox et al. (2021) performs is focused on low-velocity pulsars, and this test limits the kick velocity of an USSNe to 30 km s^{-1} , well below v_{TGRV} .

The Hobbs et al. (2005) kick has a closed-form expression for the PDF $P(v_k)$, given by Equation 5.3. Consequently, it has a smooth curve in Figure 5.5, whereas the Bray & Eldridge (2018) and Willcox et al. (2021) PDFs are presented as histograms as they are computed.

Figure 5.6 shows the K-S statistic as a function of α and β across our two parameter spaces. We also show the K-S statistic of the Hobbs kick, which has a value of 0.34. We find in general a valley of models with good agreement to Willcox et al. (2021), and find that the global minimum K-S statistic is at $\alpha = 100, \beta = -180$ – agreeing with the prior work of Bray & Eldridge (2018) (who find $\alpha = 100, \beta = -170$).

We also show in Figure 5.7 a selection of four PDFs, alongside their associated CDF and the CDFs for Willcox et al. (2021) and the Hobbs kick. We note that the numerical values shown in Figure 5.7 differs somewhat from those shown in Richards et al. (2023). This was due to a minor bug in the code incorrectly labelling the KS statistics for the figure. The actual result of the paper does not change with the correction of the bug, however the values reported in Figure 5 of Richards et al. (2023) do change.

The four values we present on the left side of Figure 5.7 correspond to $\alpha = 100 \text{ km s}^{-1}, \beta = -170 \text{ km s}^{-1}$ for the fiducial Bray kick, $\alpha = 180 \text{ km s}^{-1}, \beta = 10 \text{ km s}^{-1}$ to represent the system with the highest $\ln \mathcal{K}$ value from subsection 5.3.1, $\alpha = 170 \text{ km s}^{-1}, \beta = -10 \text{ km s}^{-1}$ the lowest, and $\alpha = 40 \text{ km s}^{-1}, \beta = 0 \text{ km s}^{-1}$ the system with low α and zero β , corresponding to a kick that would *a priori* generate logical velocities in the USSNe limit.

5.5 Production of Ultra-Stripped Supernovae

As discussed earlier, the fiducial values of the Bray kick from Bray & Eldridge (2018) admit some problematic extremal cases. If $m_{\text{ej}} \rightarrow 0$, as is the case in a USSN, the kick velocity would become negative. For instance, an “average” USSN with $m_{\text{ej}} = 0.15 M_{\odot}$ and $m_{\text{rem}} = 1.45 M_{\odot}$ would have a kick velocity of $\sim -159 \text{ km s}^{-1}$. The concern here is twofold. Firstly, a negative kick velocity would mean that the velocity vector of the kick would be antialigned with the direction of the more massive remnant; this is a violation of conservation of momentum. Secondly, a kick with high magnitude means that, irrespective of the alignment of the kick, the kick itself is strong. As discussed in Willcox et al. (2021), USSNe are associated with weak kicks, peaking at $v_k \sim 30 \text{ km s}^{-1}$. This means that the fiducial Bray kick is unable to correctly account for USSNe.

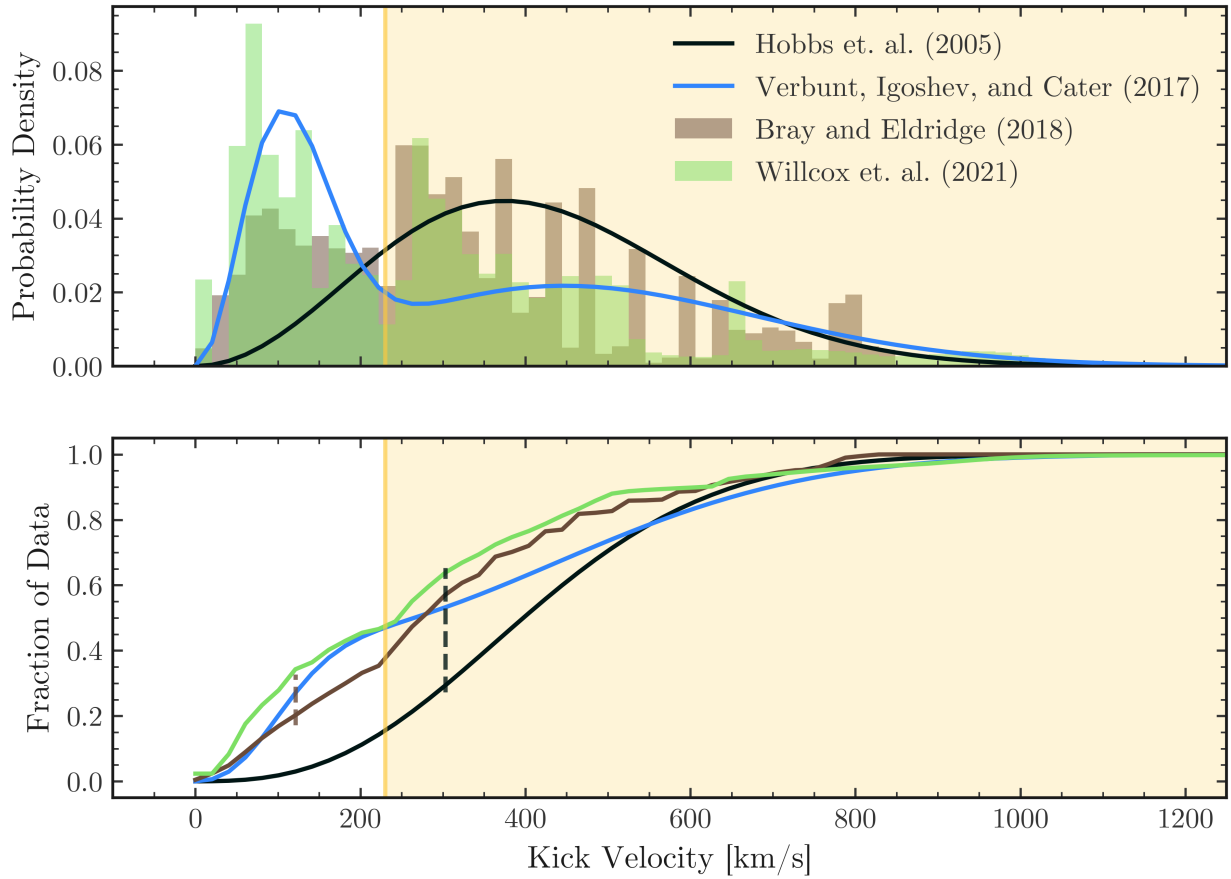


Figure 5.5: The PDF and CDF of the distributions using kicks from Hobbs et al. (2005); Verbunt et al. (2017); Bray & Eldridge (2018); Willcox et al. (2021). Hobbs et al. (2005) has a closed-form expression, whereas the others are computed empirically, hence the Bray and Willcox CDFs are less smooth than that of Hobbs. The two dashed lines indicate where and how the K-S statistic is measured for the associated distribution. The yellow region indicates the region for which $v_{\text{kick}} \geq v_{\text{TGRV}}$, the transverse Galactic rotation velocity of the Sun, $\sim 230 \text{ km s}^{-1}$.

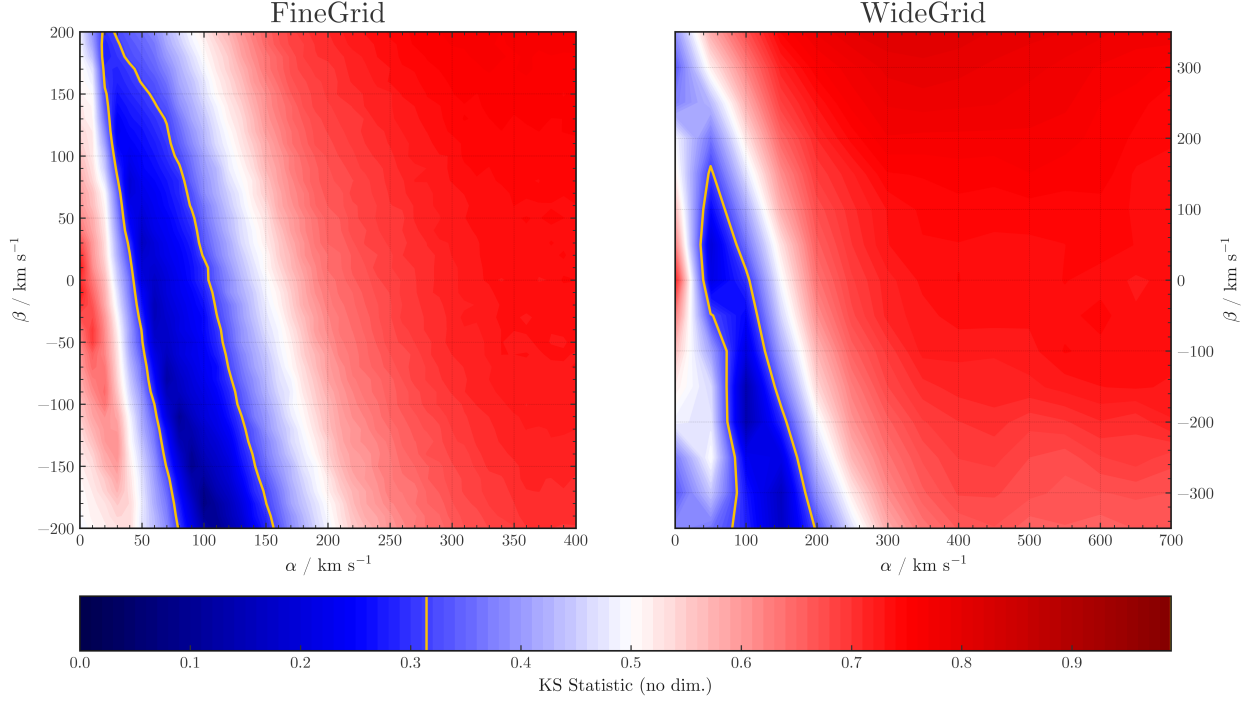


Figure 5.6: The K-S statistic for FINEGRID and WIDEGRID. The orange line denotes the K-S statistic for the Hobbs kick, which has a value of 0.34.

To account for these supernova kicks, we introduce a new constraint onto our parameters. Following Tauris et al. (2015), we define a USSN as follows:

1. The ejecta mass is in the range $0 \leq m_{\text{ej}} \leq 0.3 M_{\odot}$, admitting the USSN SN2019dge from Yao et al. (2020),
2. The remnant mass is in the range $1.1 \leq m_{\text{rem}} \leq 1.8 M_{\odot}$,
3. The envelope mass $m_{\text{env}} \leq 0.2 M_{\odot}$.

For every combination α, β , we record whether it is capable of producing one USSN. The constraint is that this heuristic is met. This constraint is extremely conservative, insofar as we do not constrain on the prevalence of USSNe in our sample, but rather by whether or not they are possible at all. The reason for the conservatism here is that the rates of Type Ic USSNe are expected to be very low. In the BPASS dataset, they comprise 1.8 per cent of all CCSNe (Stevance, 2022). The paucity of data alongside the paucity of observational evidence suggests a low rate, but since the prevalence is unknown, we place the conservative constraint of at least one USSN to be agnostic. This effectively widens our error bars such that when the USSNe rate is refined further, it will shrink the parameter space.

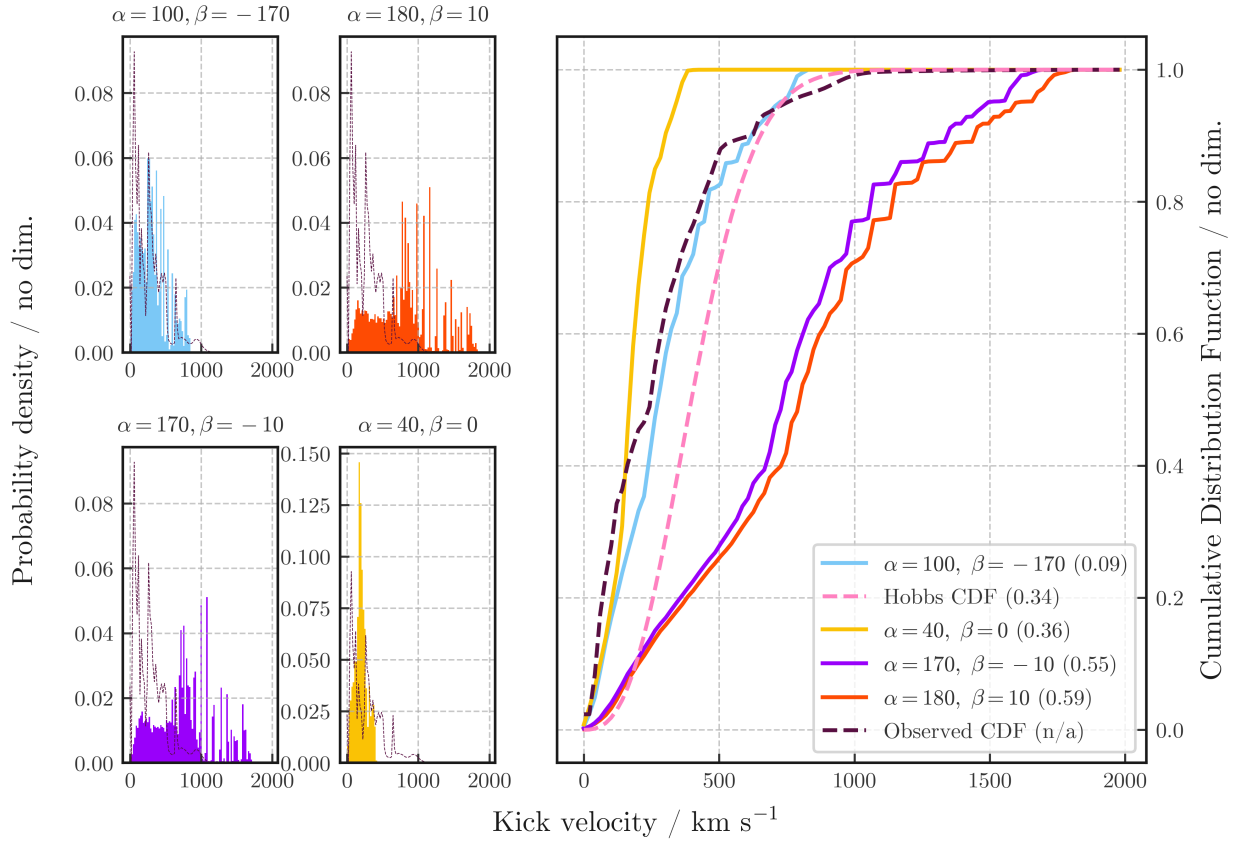


Figure 5.7: The PDFs and CDFs of four systems, alongside the Hobbs CDF and Willcox et al. (2021) CDF. The dashed line on the PDF plots is the PDF of the observed distribution from Willcox et al. (2021). On the right hand plot, the number in parentheses represents the K-S statistic for the distribution, measured relative to Willcox et al. (2021), which by definition has no K-S statistic.

We are now in a position to impose constraints on our parameter space based on the analysis provided above. We apply the following four constraints:

1. **DNS Merger Rate** (section 5.2): Does the merger rate produced by these parameters sit along the LIGO contour?
2. **DNS Period-Eccentricity Distributions** (section 5.3): Is $\ln \mathcal{K}_{\alpha\beta} \geq 3.4$ compared to the Hobbs et al. (2005) kick (representing conclusive evidence in favour of the distribution)?
3. **Isolated Pulsar Velocities** (section 5.4): Is the K-S statistic produced by this choice of parameters, in comparison to the Willcox et al. (2021) kick, less than 0.5 (indicating it agrees with observational data more than it disagrees)?
4. **Logical USSN velocities** (section 5.5): Is this choice of parameters able to produce at least one USSN?

These can be overlaid onto our parameter space, which is shown in Figure 5.8. The center of mass of our fit region is located at $\alpha = 117^{+36}_{-51} \text{ km s}^{-1}$, $\beta = 13^{+8}_{-14} \text{ km s}^{-1}$, however this value is computed on a finer grid than the parameter space and is thus likely to be an overfit to the data. We thus report the final recommended model value to the nearest half-bin-width of the parameter space, broadening error bars where necessary, and find $\alpha = 115^{+40}_{-55} \text{ km s}^{-1}$, $\beta = 15^{+10}_{-15} \text{ km s}^{-1}$ as the recommended model value. Error bars are always broadened here; traditional rounding would result in the upper bound on α (being 153 km s^{-1}) becoming 150 km s^{-1} , which artificially removes potentially permissible regions of the parameter space. Broadening the bars removes this at the cost of potentially admitting values that do not belong within the “acceptable” zone. Further constraints, however, will shrink the region, so this is an acceptable tradeoff.

In Figure 5.9, we present the two major inferred properties of our recommended model value. First, we present the CDFs of a population governed by a Bray kick with the recommended parameters, contrasted with the associated Hobbs kick and observed Willcox et al. (2021) data. We also show the period-eccentricity distribution of the population, with the observed systems from Table 5.2 overlaid atop the distribution.

We are also able to extrapolate this to comment on the expected peak of the event-rate distribution. Given that the event rate is a convolution between the merger rate of our population and the SFRD we choose (c.f. Equation 2.60), by affixing the SFRD across our populations, by Equation 2.56 we can vary $\mathcal{R}(t)$ by varying α and β . Since the SFRD is of a Gaussian form (with a peak at $z \sim 1.863$), we can also extract the peak of the distribution assuming it is governed by a particular α - β combination. The general results across the parameter space are shown in Figure 5.10. The best model choice parameters we find fall within the valley of $z = 0$, indicating that the peak of the event rate distribution is the present day.

We note that there is a single peak at $\alpha = 330 \text{ km s}^{-1}$, $\beta = -70 \text{ km s}^{-1}$ which appears to be an outlier when compared to its neighbours. We display in Figure 5.11 the local behaviour of the 9 cells around this region. One can see that this beak is bimodal with a second peak at $z = 0$, which is only 0.098 per cent lower than the reported peak, clearly resulting in a simple outlier.

Unfortunately, there is a paucity of observational evidence to confirm whether or not the peak is truly the present day. Given that a redshift of $z = 2$ corresponds to a lookback time of $t_{\text{LB}} = 10 \text{ Gyr}$, we lack

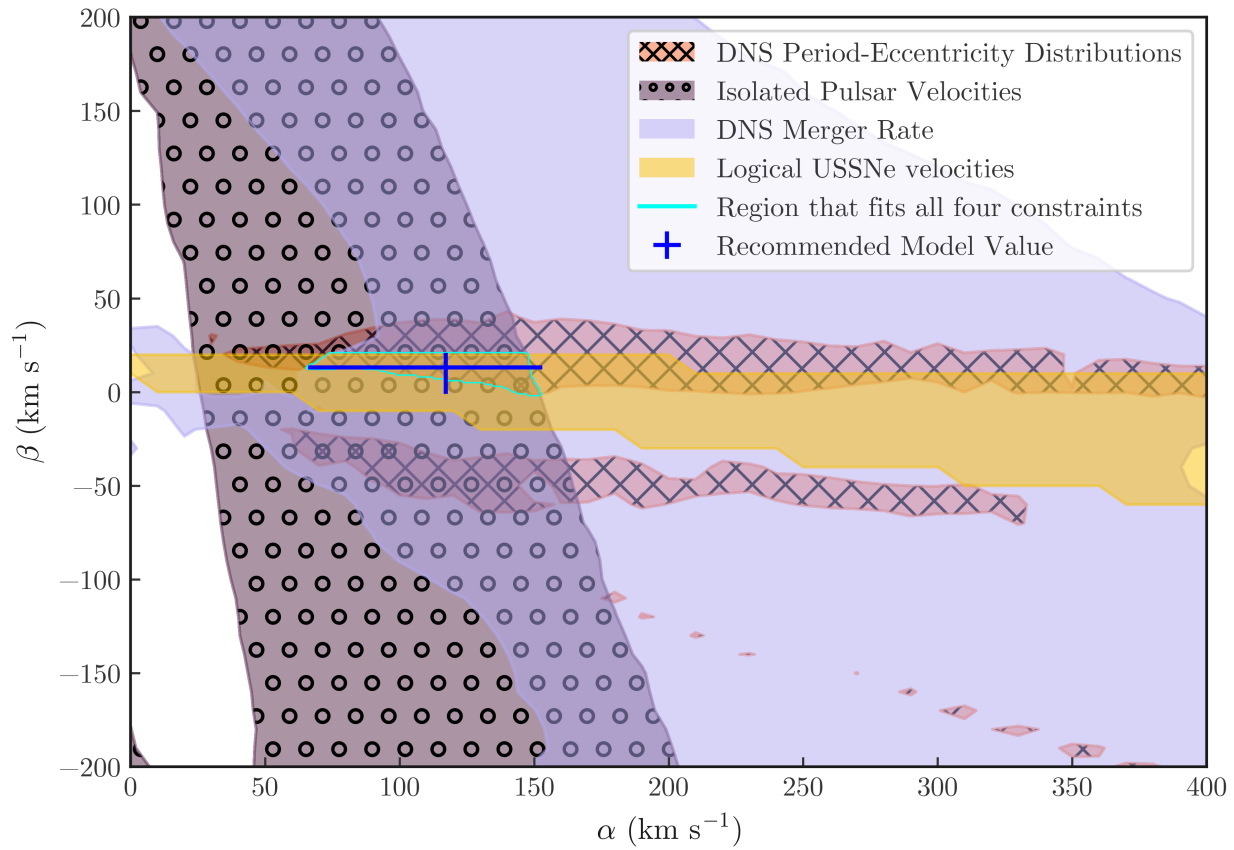


Figure 5.8: The four constraints we apply overlaid on our FINEGRID parameter space. The cyan region represents the region which fits all four parameters, and the blue cross represents the “center of mass” of this region.

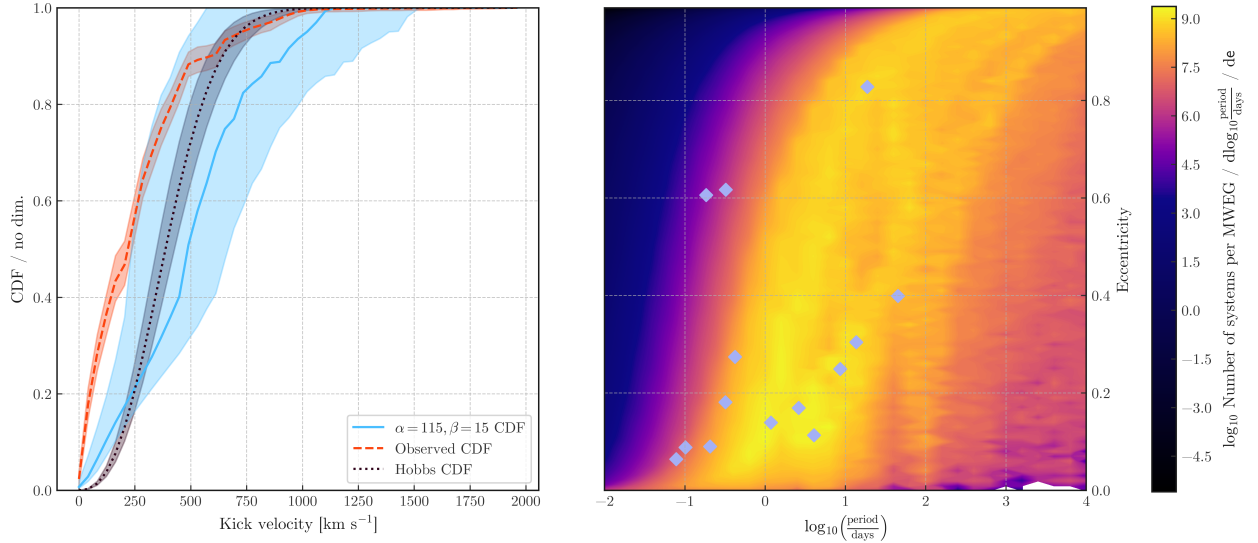


Figure 5.9: The CDF for a population of isolated pulsars (left) and the period-eccentricity distribution (right) for $\alpha = 115 \text{ km s}^{-1}, \beta = 15 \text{ km s}^{-1}$. The blue region on the CDF plot shows the uncertainty region for the reported value. The red region shows the uncertainty band on the Willcox et al. (2021) CDF, which was computed by sampling the PDF 10,000 times, taking 2,000 samples each time, and plotting the extent of the simulated distribution. The black band is the uncertainty on the Hobbs distribution, taking $\sigma_{\text{max}} = 291 \text{ km s}^{-1}$ and $\sigma_{\text{min}} = 239 \text{ km s}^{-1}$ from Verbunt et al. (2017).

evidence to corroborate this prediction. The next generation of telescopes, including JWST, may yet be able to constrain this result further. In the interim, indirect evidence may be able to be applied via the r-process enrichment history of the Milky Way. Van de Voort et al. (2022), for example, finds that greater velocity natal kicks would be correlated with fewer r-process enhanced stars in the Milky Way, as higher kicks would unbind more binaries, meaning that r-process elements are no longer able to accrete onto star-forming regions.

5.7 Chapter Summary

We have presented a formalism for evaluating and quantifying the goodness-of-fit of the natal kick mechanism against four independent tests. The tests we have presented – constraining the merger rate of BNS systems, the period-eccentricity distribution of Galactic BNS systems, the kick velocities of single-star pulsars, and the production of ultra-stripped supernovae – greatly constrain the parameter space to be less than 1 per cent of the possible parameters. Furthermore, the data suggest that the peak of the merger rate distribution is the present day, with $z \simeq 1.5$ as the upper bound of the entire parameter space, and $z = 0$ for our recommended value.

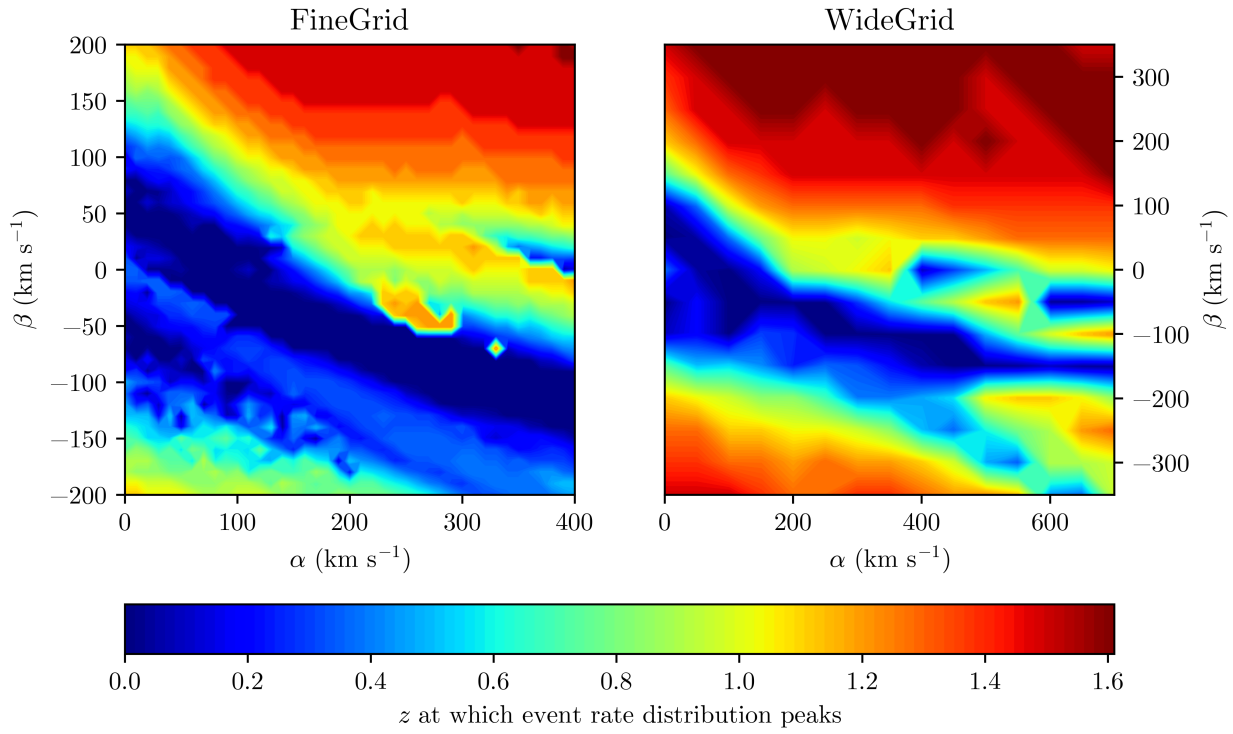


Figure 5.10: The peak of the event rate $\mathcal{R}(t)$ as a function of α and β . Varying these two parameters generally brings the peak closer to the present day ($z = 0$). Absent a merger rate (or, more formally, assuming the merger rate is instantaneous and given by a Dirac-delta function), the SFRD peaks at $z \sim 1.863$. The region surrounding the peak at $\alpha = 330, \beta = -70$ is shown in Figure 5.11.

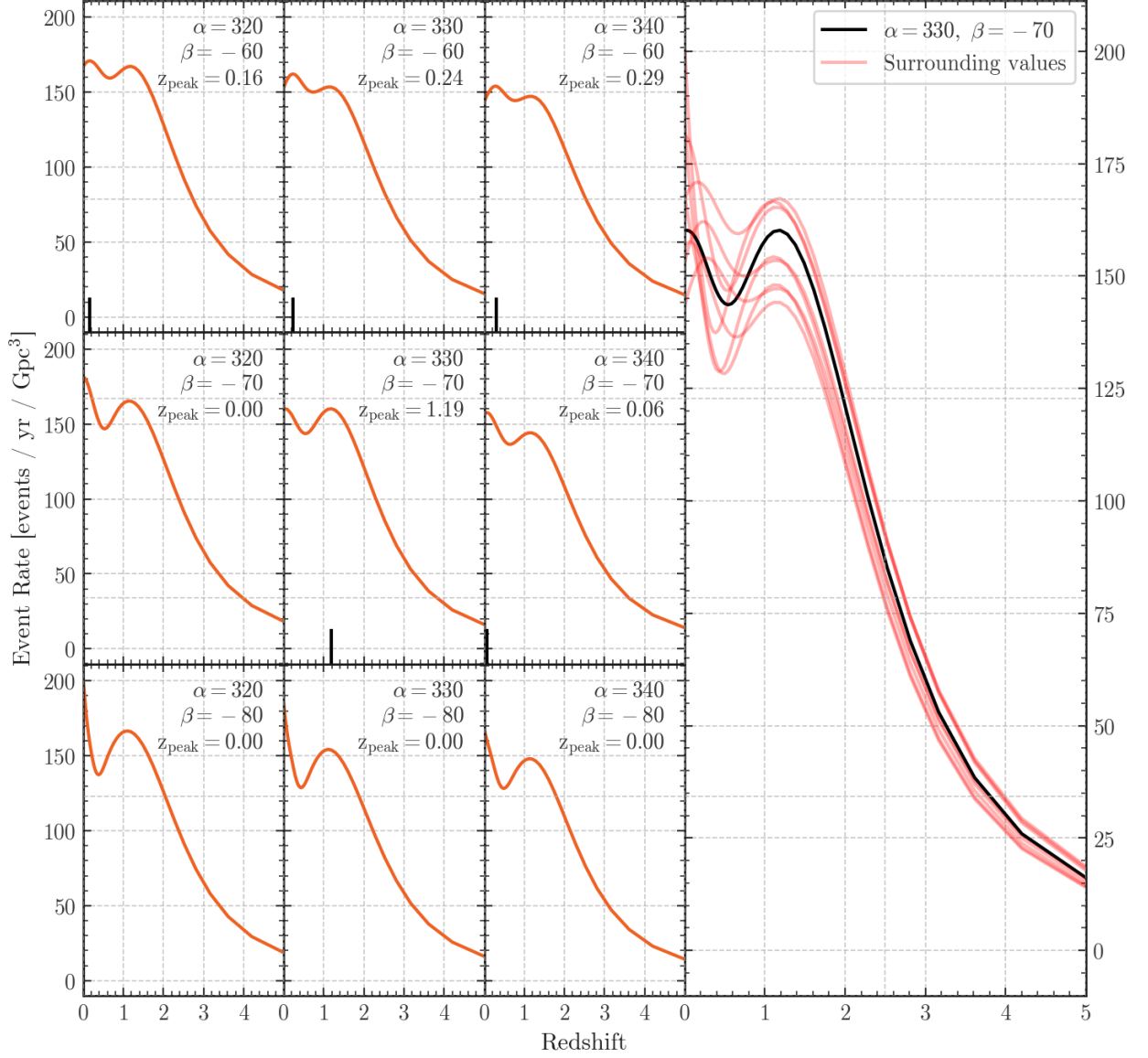


Figure 5.11: The event rate $\mathcal{R}(z)$ for a region surrounding $\alpha = 330, \beta = -70$ on FINEGRID. This region corresponds to the “outlier” in Figure 5.10. Shown on the left are the outlier bin in the center and its neighbours in both α -space and β -space. The black line on each plot denotes the location in redshift of the event rate distribution peak. The right plot shows the outlier in black and its neighbours in faded red behind.

Conclusion

*Oh! I have slipped the surly bonds of Earth
And danced the skies on laughter-silvered wings;*

...

*And, while with silent lifting mind I've trod
The high untrespassed sanctity of space,
Put out my hand, and touched the face of God.*

High Flight, BY JOHN MAGEE JR.

Take-off is optional. Landing is mandatory.

FIRST RULE OF AVIATION

6.1 Summary

6.2 Limitations

*Though in reviewing the incidents of my administration I am
unconscious of intentional error, I am nevertheless too sensible of
my defects not to think it probable that I may have committed many
errors.*

GEORGE WASHINGTON

It is a well-known remark, though attributed to no individual, that Mother Nature is indifferent to humanity. She is also indifferent to our choice of parameters – whilst we can finetune our parameter selection to approximate to essentially arbitrary precision, we have ultimately used **[TODO: ...?]**

6.3 Suggested Future Work

Appendices

Code Samples

Code Sample A.1: The TAKAHE function `integrate_eoms()`

```

1 def integrate_eoms(a0, e0, p):
2     """
3     General purpose integrator for Nyadzani & Razzaque eqns for a & e.
4
5     Params:
6         a0 - The initial semimajor axis, measured in solar radii
7         e0 - The initial eccentricity, dimensionless
8         p - A vector of parameters:
9             p[1] = m1 (units: Solar Mass)
10            p[2] = m2 (units: Solar Mass)
11            p[3] = unused (will be removed in next version)
12            p[4] = unused (will be removed in next version)
13            p[5] = Lifetime (evolution + rejuvenation)
14
15     Returns:
16         A - An array of the semimajor axes of the binary system over time. (Solar
17     ↪ Mass)
18         E - An array of the eccentricities of the binary system over time. (no
19     ↪ dim.)
20     """
21
22     Solar_Mass = 1.989e30      # kg
23     Solar_Radius = 696340000.0 # m
24     G = 6.67e-11              # m^3 kg^-1 s^-2
25     c = 299792458.0           # m/s
26
27     A = np.array([])
28     E = np.array([])
29     H = np.array([])
30
31     m1, m2, _, _, evotime = p[0], p[1], p[2], p[3], p[4]
32
33     # number of seconds in a year
34     seconds_per_year = 60 * 60 * 24 * 365.25

```

```

33
34 #####
35 #      Unit Check      #
36 #####
37 a = a0 * Solar_Radius # Meters
38 e = e0                 # Dimensionless
39 m1 = m1 * Solar_Mass  # Kilogram
40 m2 = m2 * Solar_Mass  # Kilogram
41 #####
42 #      End Unit Check  #
43 #####
44
45 # Beta has units m4 / s
46 beta = ((64/5) * G**3 * m1 * m2 * (m1 + m2) / (c**5))
47
48 A = np.append(A, a)
49 E = np.append(E, e)
50 H = np.append(H, 0.0)
51
52 total_time = 0
53
54 # Integrate until past the end of the universe, or a 10km orbit
55 while total_time/seconds_per_year + evotime < 1e11 and a > 1e4:
56     # an euler integrator: work out da/dt then times it by dt
57     # to get da, which then we can work out as a = a + da/dt * dt.
58     initial_da = (- beta / ((a**3) * (1 - e**2)**(7/2)))
59     da = initial_da * (1 + (73/24) * e**2 + (37/96) * e**4)
60
61     intial_de = (((-19/12) * beta) / (a**4*(1-e**2)**(5/2)))
62     de = intial_de * (e + (121/304) * e**3)
63     # Units: s-1
64
65     timeA = abs(1e-2 * a/da)
66
67     if e > 1e-10:
68         timeE = abs(1e-2 * e/de)
69     else:
70         de = 0
71         e = 1e-10
72         timeE = timeA * 10

```

```

73
74     # maximum timestep is half of the width of the smallest BPASS time bin
75     conv_frac = 0.23076752*0.5*seconds_per_year
76
77     dt2 = (evotime + total_time/seconds_per_year)*conv_frac
78
79     # Take a timestep that results in the smallest change: either a
80     # change in E, a change in A, 1/2 the smallest BPASS bin.
81     dt = min(timeE, timeA, dt2)
82
83     a = a + dt * da
84     e = e + dt * de
85
86     A = np.append(A, a)
87     E = np.append(E, e)
88     H = np.append(H, dt)
89
90     total_time = total_time + dt
91
92     # Determine why we stopped, so the parent function can tell what went
93     # "wrong".
94     stop_reason = "flag_not_set"
95
96     if total_time/seconds_per_year + evotime >= 1e11:
97         stop_reason = "out_of_time"
98
99     if a <= 1e4:
100         stop_reason = "merged"
101
102     # Solar Radii, Dimensionless
103     return A / Solar_Radius, E, H, stop_reason

```

Code Sample A.2: The TAKAHE function `integrate_timescale()`

```

1 def integrate_timescale(m1, m2, p0, e0):
2     """
3     General purpose integrator for eqn(5.14) of Peters, 1964 [1].
4
5     [1] https://ui.adsabs.harvard.edu/abs/1964PhRv..136.1224P/abstract
6

```

```

7  Params:
8      m1 -- The mass of the first star (Solar Mass)
9      m2 -- The mass of the second star (Solar Mass)
10     p0 -- The initial period (days)
11     e0 -- The initial eccentricity
12
13  Returns:
14     tC -- The coalescence time (seconds)
15  """
16  Solar_Mass = 1.989e30      # kg
17  Solar_Radius = 696340000.0 # m
18  G = 6.67e-11              # m3 kg-1 s-2
19  c = 299792458.0          # m/s
20
21  m1 = float(m1)
22  m2 = float(m2)
23  p0 = float(p0)
24  e0 = float(e0)
25
26  #####
27  #          BEGIN UNIT CHECK          #
28  #####
29  m1 = m1 * Solar_Mass      # Kilograms    ##
30  m2 = m2 * Solar_Mass      # Kilograms    ##
31  p0 = p0 * 24 * 60 * 60    # Seconds      ##
32  e0 = e0                   # Dimensionless ##
33  #####
34  #          END UNIT CHECK          #
35  #####
36
37  # Use Kepler's Third Law to compute the semimajor axis, in meters
38  a0 = (p0**2.0 * (G * (m1+m2))) / (4.0 * np.pi**2.0)**(1.0/3.0) # meters
39
40  # compute the constant c0
41  c0 = a0 * (1.0-e0**2.0) * e0**(-12.0/19.0) * (1.0+(121.0 / 304.0 *
42  ↪ e0**2.0))**(-870.0/2299.0) # meters
43
44  # Compute the constant beta
45  beta = 64.0 / 5.0 * G**3.0 * m1 * m2 * (m1 + m2) / c**5.0
46  # meters4 / s

```

```

46
47     # Circular binary coalescence time
48     tC = a0**4.0 / (4.0*beta)
49
50     if e0 != 0:
51         if e0 < 0.01:
52             # Low ecc - see eqn after eqn(5.14) of Peters, 1964
53             tC = c0**4.0 * e0**(48.0/19.0) / (4.0*beta)
54         elif e0 > 0.99:
55             # High ecc - see eqn after eqn after eqn(5.14) of Peters, 1964.
56             tC = tC * ((768.0 / 425.0) * (1.0-e0**2.0)**3.5)
57         else:
58             # Medium ecc - see eqn(5.14) of Peters, 1964
59             e = 0.0
60             de = e0 / 10000.0
61             summand = 0.0
62
63             while e < e0:
64                 this_integral = de * e**(29.0/19.0)
65                 this_integral *= (1.0 + (121.0/304.0) * e**2.0)**(1181.0/2299.0)
66                 this_integral /= (1.0 - e**2.0)**(1.5)
67
68                 summand += this_integral
69                 e += de
70
71                 e = float(e)
72
73             tC = (12.0/19.0) * (c0**4.0 / beta) * summand
74
75     return tC

```

[TODO: fix label before submission]

Code Sample A.3: The KAITAIKI script required to generate ??

```

1  import kaitiaki
2
3  STARS = kaitiaki.STARS.STARSController()
4  STARS.update_run_bs('.')
5
6  generate_ZAMS = False

```

```

7
8 if generate_ZAMS:
9     for target in [8, 10]:
10         STARS.blit()
11         STARS.load_default_modin()
12         STARS.setup_zams_inflation(target)
13
14         STARS.configure_parameters({
15             'DT1': 0.1,
16         })
17
18         STARS.run()
19
20         for f in ['plot', 'out', 'modout']:
21             STARS.terminal_command(f'mv {f} {f}.{target}msun')
22
23 STARS.blit()
24
25 cfigs = ['primary', 'secdry', 'binary']
26
27
28 def run_single(mode):
29     STARS.terminal_command('cp /home/sric560/Desktop/CheckNESI/data data')
30     STARS.setup_single_evolution()
31     STARS.commit_parameters()
32
33     mzams = {'primary': 10, 'secdry': 8}[mode]
34
35     STARS.modout_to_modin(modout_location=f'modout.{mzams}msun',
36                          modin_location='modin')
37
38     STARS.run()
39
40     for f in ['plot', 'out', 'modout']:
41         STARS.terminal_command(f'mv {f} {f}.{mode}')
42
43     print(f"{mode} complete.")
44
45
46 def run_binary():

```

```

47 STARS.terminal_command('cp /home/sric560/Desktop/CheckNESI/data data')
48
49 STARS.setup_binary_evolution()
50 STARS.commit_parameters()
51
52 STARS.modout_to_modin(modout_location='modout.10msun',
53                       modin_location='modin')
54
55 STARS.modout_to_modin(modout_location='modout.8msun',
56                       modin_location='modin2')
57
58 STARS.set_period(30)
59
60 STARS.run()
61
62 for f in ['plot', 'out', 'modout',
63          'plot2', 'out2', 'modout2']:
64     STARS.terminal_command(f'mv {f} {f}.binary')
65
66 print(f"Binary complete.")
67
68
69 for cfg in cfgs:
70     match cfg:
71         case 'primary' | 'secdry':
72             run_single(cfg)
73         case 'binary':
74             run_binary()

```

Code Versions

Code Name	Version	Reference
HOKI	1.7.1	Stevance et al. (2020)
KAITIAKI	1.0.0	This work
MATPLOTLIB	3.8.2	Hunter (2007)
NUMBA	0.58.1	Lam et al. (2015)
NUMPY	1.26.3	Harris et al. (2020)
PANDAS	2.1.4	McKinney (2010)
PYTHON	3.11.5	
SCIPY	1.11.4	Virtanen et al. (2020)
TAKAHE	1.0.0	Richards et al. (2023)
TUI	8	Ghodla et al. (2023)*

Table B.1: The code versions we use in this thesis. *TUI was described in earlier papers, but Ghodla et al. (2023) was the first to name it.

Period-Eccentricity distributions along the LVK Collaboration O3 contour

In Figure C.1 we display the period-eccentricity distributions along the LVK Collaboration O3 contour. We also display a white dashed line corresponding to a merger time contour of the Hubble time. Most plots have a white region in the lower right corner of the plot, which corresponds to no systems passing through those bins. We use the definition of a Milky-Way Equivalent Galaxy (MWEG) from Kasen et al. (2017), namely that there is one MWEG per $(4.4 \text{ Mpc})^3$.

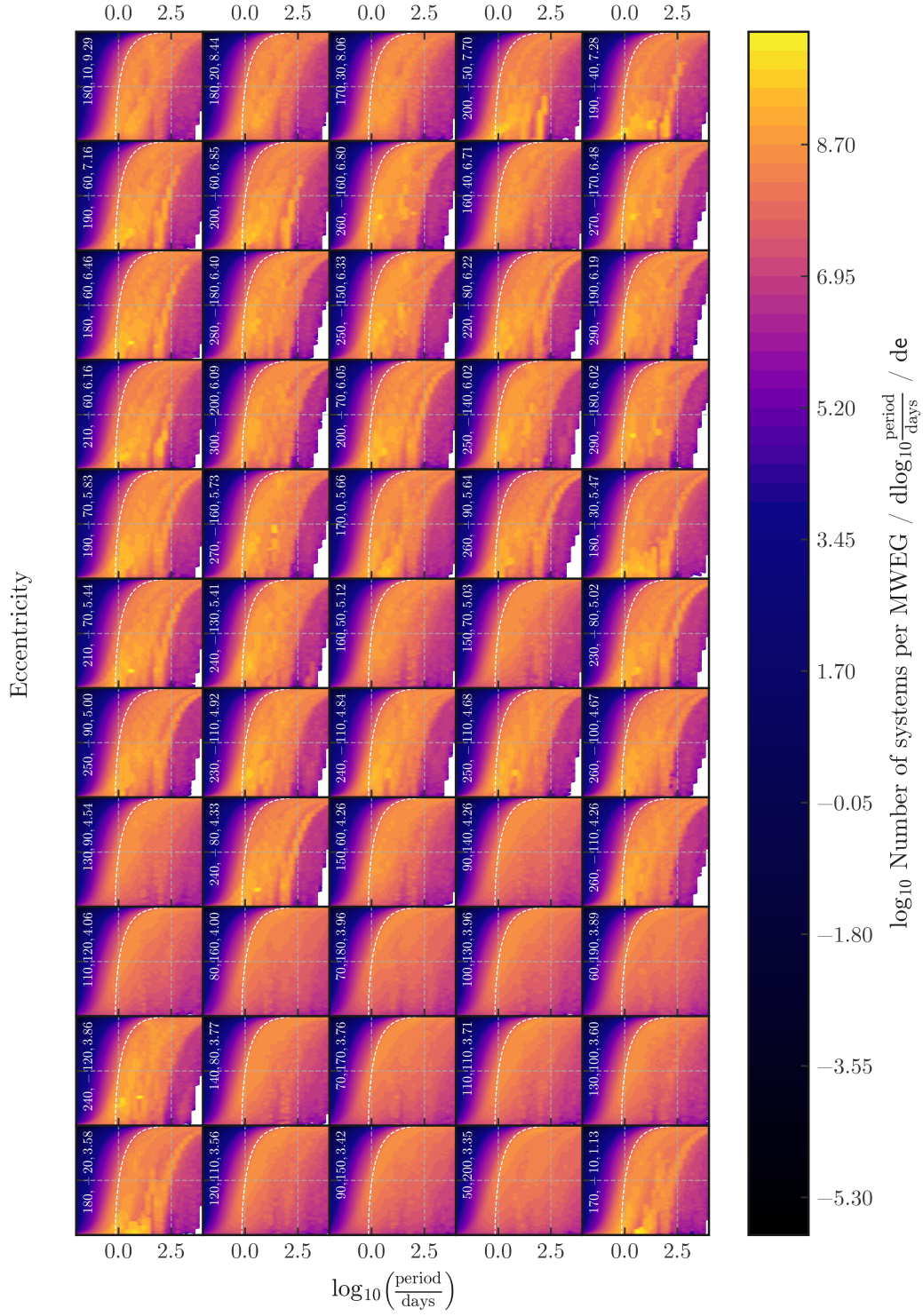


Figure C.1: The period-eccentricity distributions of populations governed by Bray kicks across the LIGO O3 contour. The combination of α and β used in the kick prescription is given in white along the left side of each plot. White regions denote no systems passing through those bins.

Parameter Configuration

In this appendix we list the default parameters chosen for our binary evolution runs, as detailed in subsection 2.7.1.

Parameter	Value	Comment	Parameter	Value	Comment	Parameter	Value	Comment
NM2	199		ICL	1		IT1	40	
ION	2		IT2	20		IAM	1	
JIN	15		IOP	5		OUT	15	
INUC	0		NCH	3		IBC	0	
JP	1		ICN	0		ITH	1	
IML1	5		IX	1		IML2	5	
IY	1		ISGTH	0		IZ	1	
IMO	0		IMODE	2		IDIFF	0	
NWRT1	100		EPS	10^{-6}		NWRT2	1	
DEL	0.01		NWRT3	2		DHO	10^{-8}	
NWRT4	1		DT3	0		NWRT5	0	
DDD	1.0		NSAVE	10		NMONT	0	
DT1	0.1		DT2	1.2		CT1	3.0	
CT2	3.0		CT3	0.05		CT4	0.01	
CT5	0.6		CT6	0.35		CT7	0.45	
CT8	10^{-4}		CT9	0.1		CT10	0.3	

Table D.1: Our general initial conditions. Unless otherwise described, these are assumed to be enabled for all runs.

Parameter	Value	Comment	Parameter	Value	Comment	Parameter	Value	Comment
ZS	0.02		ALPHA	1.925		CH	0.7	
CC	0.173		CN	0.053		CO	0.482	
CNE	0.099		CMG	0.038		CSI	0.08	
CFE	0.072		RCD	10 ⁶		OS	0.12	
RML	0		RMG	0		ECA	0	
XF	0.1		DR	0.001		RMT	1000.0	
RHL	0		AC	0		AK1	1.0	
AK2	10 ⁻⁴		ECT	0		TRB	0	
IRAM	1		IRS1	1		VROT1	0	
IRS2	1		VROT2	10.0		FMAC	1.0	
FAM	0		IVMC	0		TRC1	700.0	
IVMS	0		TRC2	0		MWTS	0	
IAGB	0		ISGFAC	0		FACSGMIN	0.01	
SGTHFAC	0		ISTART	0		HKH	1.0	
GFF	10.0		NNMOD	0		CEPR	0	
ALPHACE	1.0		-	-	-	-	-	-

Table D.1: Continued

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ISA LEI – NANOQU RARAWA,
Miko sana vodo e na mataka,
Bau nonuma, na nodatau lasa,
Mai Suva, nonuma tikoga.

ISA LEI. THE PURPLE SHADOWS FALL,
Sad the morrow will dawn upon my sorrow.
Ah, forget not when you are far away
Precious moments beside dear Suva Bay.

– *Isa Lei*, “The Fijian Farewell Song”